ENGLISH FOR WORK / METRICS / QUALITY / MODELS / DECISIONS

Data Analytics & Business Intelligence English

Define the measure. Make the finding useful.



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Eight practical lessons. Eleven extended conversations.
Vocabulary, phrase banks, guided speaking, and explained answers.

Revised 2026 edition / Upper-intermediate to advanced

Your route through the book

01 Metric Definitions and Business Questions, page 4

Reconcile apparently conflicting metrics by stating the event, distinct entity, window, and denominator.

02 Data Quality and Trust, page 16

Explain mixed source freshness and set an appropriate reporting boundary without mislabeling missing data as zero.

03 Dashboards and Executive Reporting, page 29

Turn an isolated monthly count into a clear performance statement with definition, target, comparison, and units.

04 Causation, Correlation, and Caveats, page 41

Report a meaningful association while separating the observed difference from an unestablished causal effect.

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06 Experiment Readouts and Statistical Thinking, page 67

Present an experiment estimate with its interval, practical threshold, and pre-agreed decision rule.

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Specify a purpose-limited data request, distinguish pseudonymization from anonymity, and preserve approval and retention controls.

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Suggested study range: B2-C1. This is an editorial study range, not a certified proficiency assessment.

A number needs a definition and a purpose.

Analytics work often fails in the conversation after the calculation. Teams may use one label for different events, confuse a refresh time with source coverage, or present a promising estimate as a guaranteed effect. These cases supply the facts and practice the language that keeps a useful conclusion attached to its actual evidence.

Agree what is being counted

Name the entity, event, window, and exclusions. A familiar metric label is not a complete definition.
"We counted distinct users with a completed transaction during September."

Put the denominator into speech

A percentage needs an eligible total. Say whether a change is relative or in percentage points.
"That is 480 of 500 cases, or 96%, one point above target."

Keep the limitation near the claim

Qualify source coverage and uncertainty where the result is reported, not only in an appendix.
"The dashboard refreshed today; refunds are complete only through Monday."

End with a bounded recommendation

Connect the finding to an action, owner, and review point without claiming that the cause is already known.
"Approve a two-day investigation of the documented service issue."

> ### Professional boundaries
>
> Original fictional English practice, not statistical certification, legal advice, a privacy determination, or authorization to use customer data. All datasets, thresholds, local approval rules, and business decisions are invented. Actual analysis must follow the applicable definitions, methods, access controls, and responsible review.

A Read the situation

LESSON 01 / METRIC DEFINITIONS AND BUSINESS QUESTIONS

Reconcile apparently conflicting metrics by stating the event, distinct entity, window, and denominator.

Two active-user totals, two different events

Analyst Mei and product lead Rafael compare September figures for the same 2,000 eligible users, using UTC calendar boundaries and excluding internal and test accounts. One report counts 1,200 distinct users with a successful login. Another counts 900 distinct users with a completed transaction. All 900 are within the login group. Both queries match their stated definitions. The business question is how many eligible users completed a transaction, not how many signed in. The team must choose an accurate label and rate.

People: Mei, Data analyst; Rafael, Product lead.

Read the record

Choose one answer per question using the facts above.

1. Which count answers the supplied business question?

A. 900 distinct users with a completed transaction
B. 1,200 distinct users with a login
C. 2,100 users obtained by adding the groups
D. 2,000 completed transactions

2. What is the correct eligible-user transaction rate?

A. 75%
B. 105%
C. 45%
D. 60%

3. Why can both query results be correct?

A. Every metric must have the same count.
B. The larger number is always more accurate.
C. A login and a transaction are identical events.
D. They apply different event definitions to the same eligible population.

Answers and explanations, page 121

B Vocabulary in use

CORE TERMS

metric
A defined quantitative measure.

define a metric

metric definition
Rules specifying exactly how a measure is calculated.

agree a metric definition

distinct user
A unique user counted once under a defined identity rule.

count distinct users

eligible population
The complete group allowed into the measure's base.

define the eligible population

numerator
The counted outcome placed over the base of a rate.

verify the numerator

calendar month
A month bounded by the adopted calendar and time zone.

report by calendar month

business question
The decision-relevant question an analysis is meant to answer.

clarify the business question

qualifying event
An action meeting the condition for inclusion.

specify the qualifying event

event count
The number of recorded qualifying occurrences.

distinguish event counts from users

denominator
The total used as the base of a rate.

state the denominator

reporting window
The time period included in a measure.

align reporting windows

time zone
The time reference used to assign event dates.

fix the reporting time zone

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

metric METRIC

define a metric

Keep the meaning precise: A defined quantitative measure.

event count EVENT COUNT

distinguish event counts from users

Keep the meaning precise: The number of recorded qualifying occurrences.

B Precision and professional shorthand

EXTEND YOUR RANGE

exclusion rule
A condition removing records from the measure.
document exclusion rules

internal account
An account belonging to the organization's own personnel or systems.
exclude internal accounts

test account
An account used for testing rather than genuine activity.
filter test accounts

deduplication
Removing repeated representations of the same counted entity.
apply user deduplication

identity resolution
Connecting records that refer to the same entity.
review identity resolution

active user
A user meeting a specified activity definition.
define active users explicitly

login-based activity
Activity defined by a qualifying sign-in.
report login-based activity

transaction-based activity
Activity defined by a qualifying transaction.
report transaction-based activity

subset
A group entirely contained within another group.
identify the subset relationship

metric owner
The role accountable for a measure's definition and use.
identify the metric owner

semantic layer
Shared definitions connecting data to business meaning.
maintain a semantic layer

reconciliation
Explaining and resolving differences between reported figures.
reconcile metric totals

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

time zone TIME ZONE

fix the reporting time zone

Keep the meaning precise: The time reference used to assign event dates.

identity resolution IDENTITY RESOLUTION

review identity resolution

Keep the meaning precise: Connecting records that refer to the same entity.

C Phrases that move the work forward

Clarify the question
"Are we measuring sign-in or completed transactions?"

Define the counted entity
"Each user is counted once."

Name the event
"The qualifying event is a completed transaction."

Set the window
"Use the September calendar month in UTC."

Keep exclusions aligned
"Both measures exclude internal and test accounts."

Acknowledge both results
"Both queries match their own definitions."

Explain the difference
"The reports use different activity events."

Choose the relevant count
"Nine hundred users answer the transaction question."

State the base
"The eligible population is 2,000 users."

Give the rate
"That is 45% of eligible users."

Distinguish another rate
"Seventy-five percent uses logged-in users as its base."

Avoid double-counting
"The transaction group is a subset of the login group."

Improve the label
"Call this transaction-active users."

Preserve the other measure
"Keep the login measure under its own label."

Assign definition ownership
"The metric owner should approve the shared definition."

Close the disagreement
"We need consistent meaning, not an artificial match between different events."

Precision in real exchanges

Purpose: Set the window
Useful wording: "Use the September calendar month in UTC."

Purpose: Keep exclusions aligned
Useful wording: "Both measures exclude internal and test accounts."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which label best fits the 900 count?

A. September transaction-active users
B. All September logins
C. Completed transaction events, including repeats
D. Total eligible accounts

2. What does 900/1,200 measure?

A. A 75% growth rate from last month
B. The share of logged-in users who completed a transaction
C. The share of all eligible users who logged in
D. The number of repeated transactions

3. A colleague needs the term for this meaning: An action meeting the condition for inclusion. Which term fits?

A. test account
B. numerator
C. qualifying event
D. subset

D Choose the response that fits / continued

4. Rebuild the exchange from "Two active-user totals, two different events". Rafael: "The window is the September calendar month in UTC for both, and they use the same two thousand eligible users. That part is aligned." Mei: [missing response] Rafael: "For the actual question, I should choose nine hundred and label it as people completing a transaction. Multiple transactions by one user still count once." Which response belongs in the gap?

A. That uses a different denominator. Nine hundred divided by two thousand eligible users is forty-five percent. Nine hundred divided by twelve hundred logged-in users is seventy-five percent.

B. Good. A shared reporting window removes one possible source of disagreement. The distinction here is activity type, not different month boundaries or different treatment of test accounts.

C. Assign a metric owner and record the approved version. That gives both teams a place to resolve changes before comparing totals that no longer mean the same thing.

D. That completes the reconciliation. We have preserved both correct calculations, selected the measure that answers the decision, and removed the ambiguity that created the apparent conflict.

Answers and explanations, page 121

One useful distinction

900/2,000 is 45% of eligible users. 900/1,200 is 75% of users who logged in. Both calculations are valid for their stated bases, but only the first answers the supplied eligible-user question.

E Conversation

Two active-user totals, two different events

Two active-user totals, two different events

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

business question / denominator / reconciliation / numerator / metric owner / qualifying event / subset / distinct user / reporting window / semantic layer

01 **Rafael:** The executive slide says twelve hundred active users, but Finance has nine hundred. Both teams insist their query is right. Which number should I remove?

02 **Mei:** First agree the [1] _____. We need the number of eligible users who completed a transaction during September, not merely the number who signed in.

03 **Rafael:** The twelve hundred report counts successful logins, once per user. Finance counts completed transactions, also once per user. Both exclude internal and test accounts.

04 **Mei:** Then the [2] _____ differs. The queries can both be correct while answering different questions. We should not change a valid calculation simply to make two different meanings agree.

05 **Rafael:** The window is the September calendar month in UTC for both, and they use the same two thousand eligible users. That part is aligned.

06 **Mei:** Good. A shared [3] _____ removes one possible source of disagreement. The distinction here is activity type, not different month boundaries or different treatment of test accounts.

E Conversation / continued

07 **Rafael:** For the actual question, I should choose nine hundred and label it as people completing a transaction. Multiple transactions by one user still count once.

08 **Mei:** Yes, the entity is a [4] _____. That is not the total number of transaction events. The slide should make the counted entity clear enough that the audience cannot silently substitute another.

09 **Rafael:** I have seventy-five percent on the slide: nine hundred divided by twelve hundred. Is that answering the eligible-user question, or have I changed the base?

10 **Mei:** That uses a different [5] _____. Nine hundred divided by two thousand eligible users is forty-five percent. Nine hundred divided by twelve hundred logged-in users is seventy-five percent.

Conversation continues on the next page.

Two active-user totals, two different events

CONVERSATION / CONTINUED

11 **Rafael:** Both rates tell us something, but forty-five percent is the answer to the eligible-user question. Seventy-five percent describes the logged-in group instead.

12 **Mei:** Exactly. Keep the [6] _____ and base together in speech. Saying just the percentage would hide the change in meaning and invite an incorrect comparison later.

13 **Rafael:** Someone suggested adding the two reports to show total engagement. But the nine hundred transacting users are already included in the twelve hundred who logged in.

14 **Mei:** That [7] _____ relationship makes addition inappropriate for distinct people. It would count the transaction group twice rather than identify twenty-one hundred different active users.

15 **Rafael:** I will keep the login figure because it is useful, but give it a different label. The executive headline will use the transaction definition.

16 **Mei:** Put both in the [8] _____ with the event, identity rule, exclusions, and time zone. Renaming one slide is not enough if other reports keep reusing the ambiguous active-user label.

17 **Rafael:** Who signs off if the transaction event changes next month? I do not want the same label to hide a new definition.

18 **Mei:** Assign a [9] _____ and record the approved version. That gives both teams a place to resolve changes before comparing totals that no longer mean the same thing.

Two active-user totals, two different events / continued

19 **Rafael:** My summary will be nine hundred transaction-active users, forty-five percent of two thousand eligible users, September UTC. The login measure remains separately labeled.

20 **Mei:** That completes the [10] _____. We have preserved both correct calculations, selected the measure that answers the decision, and removed the ambiguity that created the apparent conflict.

CHECK THE GAPS

1. business question / **2.** qualifying event / **3.** reporting window / **4.** distinct user / **5.** denominator / **6.** numerator / **7.** subset / **8.** semantic layer / **9.** metric owner / **10.** reconciliation

Why these answers fit, page 123

F Check accuracy and scope

Two active-user totals, two different events

Return to the case

Analyst Mei and product lead Rafael compare September figures for the same 2,000 eligible users, using UTC calendar boundaries and excluding internal and test accounts. One report counts 1,200 distinct users with a successful login. Another counts 900 distinct users with a completed transaction. All 900 are within the login group. Both queries match their stated definitions. The business question is how many eligible users completed a transaction, not how many signed in. The team must choose an accurate label and rate.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why should the two totals not be added?

A. Each report contains exactly the same 1,200 people.
B. The 900 transacting users are already included among the 1,200 login users.
C. All additions of user counts are invalid.
D. The UTC boundary changes addition rules.

2. Which next step addresses the real disagreement?

A. Alter a correct query until both totals match.
B. Use whichever number looks more impressive.
C. Remove the denominator from the slide.
D. Record both event definitions and select the transaction measure for this question.

Answers and explanations, page 124

Listen for the distinction

Switch roles for turns 11-20. Keep the subset relationship clear when rejecting addition.

Treat a disagreement as a definition check

When two teams defend different totals, ask what each number counts before deciding which team made an error. Preserve useful measures with accurate names. A measure can be correctly calculated yet poorly suited to the business question on the table.

F Say it / transfer to a new case

One count, two possible bases

A month has 1,000 eligible users, 800 distinct login users, and 400 distinct transacting users, all within the login group. The question asks for transaction activity among all eligible users.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

40 / 400 / 1000 / 50

01 **Analyst:** The relevant distinct-user count is [1] _____.

02 **Manager:** The eligible-user denominator is [2] _____.

03 **Analyst:** The requested rate is [3] _____ percent.

04 **Manager:** Using login users instead would give [4] _____ percent.

CHECK THE GAPS

1. 400 / **2.** 1000 / **3.** 40 / **4.** 50

Why these answers fit, page 124

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Two active-user totals, two different events". Rafael: "Someone suggested adding the two reports to show total engagement. But the nine hundred transacting users are already included in the twelve hundred who logged in." Mei: [missing response] Rafael: "I will keep the login figure because it is useful, but give it a different label. The executive headline will use the transaction definition." Which response belongs in the gap?

A. First agree the business question. We need the number of eligible users who completed a transaction during September, not merely the number who signed in.

B. Exactly. Keep the numerator and base together in speech. Saying just the percentage would hide the change in meaning and invite an incorrect comparison later.

C. That subset relationship makes addition inappropriate for distinct people. It would count the transaction group twice rather than identify twenty-one hundred different active users.

D. That completes the reconciliation. We have preserved both correct calculations, selected the measure that answers the decision, and removed the ambiguity that created the apparent conflict.

Answers and explanations, page 125

A Read the situation

LESSON 02 / DATA QUALITY AND TRUST

Explain mixed source freshness and set an appropriate reporting boundary without mislabeling missing data as zero.

> **Refreshed today, complete only through Monday**
>
> Reporting analyst Asha reviews a dashboard with operations manager Tomas on Wednesday at 09:00 UTC. The dashboard refresh completed, and the orders source is verified complete through Tuesday 23:59 UTC. The refunds source is verified complete only through Monday 23:59 UTC because its next delivery is delayed. A Monday-Tuesday net-order-value figure subtracts available refunds from orders, so the periods are misaligned. Asha can publish a verified Monday-only comparison; Tuesday refunds are unknown, not confirmed zero. No delivery time for the missing source is confirmed.

People: Asha, Reporting analyst; Tomas, Operations manager.

Read the record

Choose one answer per question using the facts above.

1. Which source is complete only through Monday?

A. Refunds
B. Orders
C. Both sources through Tuesday
D. Neither source through any date

2. What is known about Tuesday refunds?

A. They are confirmed zero.
B. They equal Tuesday orders.
C. They were all processed on Monday.
D. They are unknown in the available source.

3. Which comparison can Asha publish as verified?

A. A Wednesday full-day total
B. A guaranteed delivery time for the missing source
C. Monday-only, using aligned source coverage
D. Monday-Tuesday net value without qualification

Answers and explanations, page 126

B Vocabulary in use

CORE TERMS

data quality
Fitness of data for its intended use.
assess data quality

freshness
How recent data is relative to the intended use.
monitor data freshness

timeliness
Availability of data within the required time.
define timeliness requirements

completeness
Presence of the required records or values.
verify source completeness

accuracy
Agreement between data and the facts it represents.
assess data accuracy

validity
Conformance to defined formats or rules.
test data validity

consistency
Agreement across related representations or rules.
check data consistency

uniqueness
Absence of unintended duplicate records.
test record uniqueness

refresh timestamp
The time a report or model was last updated.
display the refresh timestamp

source cutoff
The boundary through which source data is available or complete.
state the source cutoff

event time
When the represented business event occurred.
distinguish event time from load time

load time
When data entered the relevant system.
record load time

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

data quality DATA QUALITY
assess data quality
Keep the meaning precise: Fitness of data for its intended use.

validity VALIDITY
test data validity
Keep the meaning precise: Conformance to defined formats or rules.

B Precision and professional shorthand

EXTEND YOUR RANGE

upstream source
An earlier system supplying data to a pipeline.
inspect upstream sources

pipeline
A sequence of data movement and transformation steps.
monitor the data pipeline

ingestion delay
A lag in receiving data from a source.
investigate ingestion delay

late-arriving data
Records received after the expected processing point.
handle late-arriving data

backfill
Loading previously missing historical records.
validate a backfill

watermark
A tracked boundary used to process or assess data progress.
inspect the pipeline watermark

coverage window
The period or population represented by available data.
align coverage windows

reconciliation check
A comparison used to verify agreement between records or totals.
run reconciliation checks

null value
A marker for missing or unknown information in a data field.
distinguish null from zero

zero value
A recorded quantity of none, not automatically a missing value.
preserve zero values correctly

data-quality notice
A visible explanation of a known data limitation.
publish a data-quality notice

recovery estimate
A qualified expectation for restoration of a data flow.
qualify the recovery estimate

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

load time LOAD TIME

record load time

Keep the meaning precise: When data entered the relevant system.

backfill BACKFILL

validate a backfill

Keep the meaning precise: Loading previously missing historical records.

C Phrases that move the work forward

Acknowledge the refresh

"The dashboard refresh completed at 09:00 UTC."

Name the lagging source

"Refunds are complete only through Monday."

Locate the affected measure

"The two-day net figure mixes coverage periods."

Offer a usable view

"We can publish the verified Monday-only comparison."

Avoid a false promise

"The missing delivery time is not confirmed."

Describe the dependency

"The calculation depends on both orders and refunds."

Preserve the old status

"Keep the affected view marked incomplete until validation passes."

Bound the current claim

"This view is complete through Monday only."

Separate source coverage

"That does not mean every source is current through Tuesday."

State the other cutoff

"Orders are verified through Tuesday 23:59 UTC."

Distinguish missing from zero

"Tuesday refunds are unknown, not confirmed zero."

Keep the limitation visible

"Place the source-cutoff note beside the figure."

Ask for the right evidence

"Check source completeness, not only job completion."

Name the next validation

"Reconcile the backfill before republishing the two-day figure."

Clarify timing

"Load time and event time are different."

Close the update

"Publish the aligned period and report the remaining source delay."

Precision in real exchanges

Purpose: State the other cutoff
Useful wording: "Orders are verified through Tuesday 23:59 UTC."

Purpose: Locate the affected measure
Useful wording: "The two-day net figure mixes coverage periods."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which dashboard note is accurate?

A. Refreshed Wednesday 09:00 UTC; refunds complete through Monday; two-day net value incomplete.

B. Refreshed Wednesday, so all source events are current.

C. Tuesday refunds are zero because none arrived.

D. The refund delay proves every order is inaccurate.

2. Why is Monday-only reporting usable here?

A. Monday is always more accurate than Tuesday.

B. Removing refunds would make any period complete.

C. A short period never needs validation.

D. Both sources are verified complete through that shared period.

3. A colleague needs the term for this meaning: Presence of the required records or values. Which term fits?

A. null value

B. completeness

C. refresh timestamp

D. ingestion delay

D Choose the response that fits / continued

4. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "Then the two-day net figure has Tuesday orders but not the corresponding Tuesday refunds. It could look better simply because that input is incomplete." Asha: [missing response] Tomas: "The chart fills the empty Tuesday cell with zero. Should we change that to unknown while the refund delivery is missing?" Which response belongs in the gap?

A. Read the source cutoff. Refunds are complete through Monday at twenty-three fifty-nine UTC. The next delivery is delayed, so Tuesday is not covered by the available records.

B. Yes. The coverage window is misaligned. We cannot interpret the result as a complete two-day net amount when the subtraction uses refunds from a shorter period.

C. Not yet. The refresh timestamp is Wednesday at nine UTC, but the inputs have different coverage. Orders are verified through Tuesday; refunds are complete only through Monday.

D. Yes. That respects completeness for the shared period in this case. We can publish the Monday-only comparison and make clear that the two-day figure remains incomplete.

Answers and explanations, page 126

One useful distinction

A refresh timestamp records when processing ran, not how current every input was. Even a latest event timestamp alone would not prove completeness. Here, the brief explicitly verifies different complete-through cutoffs for the two sources.

E Conversation

Refreshed today, complete only through Monday

Refreshed today, complete only through Monday

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

recovery estimate / refresh timestamp / coverage window / data-quality notice / completeness / zero value / event time / backfill / source cutoff / timeliness

01 **Tomas:** The dashboard says refreshed this morning. Can I use the Monday-Tuesday net-order-value number in the operations meeting as our complete two-day result?

02 **Asha:** Not yet. The [1] _____ is Wednesday at nine UTC, but the inputs have different coverage. Orders are verified through Tuesday; refunds are complete only through Monday.

03 **Tomas:** I had assumed that a successful refresh meant every table contained the latest business activity. Which boundary should I read instead for the refund data?

04 **Asha:** Read the [2] _____. Refunds are complete through Monday at twenty-three fifty-nine UTC. The next delivery is delayed, so Tuesday is not covered by the available records.

05 **Tomas:** Then the two-day net figure has Tuesday orders but not the corresponding Tuesday refunds. It could look better simply because that input is incomplete.

06 **Asha:** Yes. The [3] _____ is misaligned. We cannot interpret the result as a complete two-day net amount when the subtraction uses refunds from a shorter period.

E Conversation / continued

07 **Tomas:** The chart fills the empty Tuesday cell with zero. Should we change that to unknown while the refund delivery is missing?

08 **Asha:** Missing is not a [4] _____. We do not know the amount yet. Showing zero as the business result would turn a source limitation into a false claim about customer activity.

09 **Tomas:** I still need a reliable number for the meeting. Could we show Monday by itself, since both sources are verified complete through that day?

10 **Asha:** Yes. That respects [5] _____ for the shared period in this case. We can publish the Monday-only comparison and make clear that the two-day figure remains incomplete.

Conversation continues on the next page.

Refreshed today, complete only through Monday

CONVERSATION / CONTINUED

11 **Tomas:** The note should be near the result, not somewhere people will miss it. I will explain that Tuesday orders alone do not settle the net figure.

12 **Asha:** Add a [6] _____ beside the affected view. State which source is delayed, its complete-through time, and which period remains usable rather than labeling the entire dashboard unreliable.

13 **Tomas:** Do we know when the missing refund delivery will arrive? I would like to tell the meeting whether the full figure will be available this afternoon.

14 **Asha:** There is no confirmed [7] _____. I will request an update from the source owner, but I should not invent a delivery time simply because the meeting needs one.

15 **Tomas:** Once the rows arrive, we will have business events from Tuesday loaded on a later day. We should not move those refunds into Wednesday totals.

16 **Asha:** Correct. Preserve [8] _____ separately from load time. The records belong to the appropriate business period even when the pipeline receives them later than expected.

17 **Tomas:** When the delivery arrives, who checks that it is complete? I do not want the warning removed just because the refresh turns green.

18 **Asha:** Validate the [9] _____ and document the checks. A green refresh icon should not be our only evidence that the previously incomplete reporting period is now fit for use.

19 **Tomas:** I will take the Monday-only view to the meeting, explain the Tuesday refund gap, and avoid promising a two-day net result before it is verified.

20 **Asha:** That is the right [10] _____ message: a useful aligned result now, a specific limitation, and a clear validation step before the fuller figure is released.

CHECK THE GAPS

1. refresh timestamp / **2.** source cutoff / **3.** coverage window / **4.** zero value / **5.** completeness / **6.** data-quality notice / **7.** recovery estimate / **8.** event time / **9.** backfill / **10.** timeliness

Why these answers fit, page 127

F Check accuracy and scope

Refreshed today, complete only through Monday

Return to the case

Reporting analyst Asha reviews a dashboard with operations manager Tomas on Wednesday at 09:00 UTC. The dashboard refresh completed, and the orders source is verified complete through Tuesday 23:59 UTC. The refunds source is verified complete only through Monday 23:59 UTC because its next delivery is delayed. A Monday-Tuesday net-order-value figure subtracts available refunds from orders, so the periods are misaligned. Asha can publish a verified Monday-only comparison; Tuesday refunds are unknown, not confirmed zero. No delivery time for the missing source is confirmed.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which check is insufficient by itself?

A. Seeing that the dashboard refresh job completed
B. Verifying source coverage against expected records
C. Reconciling the missing source after backfill
D. Checking the complete-through cutoff for each input

2. What should happen after the delayed records arrive?

A. Validate and reconcile them before removing the limitation.
B. Immediately claim that every historical figure was already correct.
C. Keep treating the records as zero.
D. Change the event dates to Wednesday load time.

Answers and explanations, page 129

Listen for the distinction

Switch roles for turns 11-20. Stress event time rather than the later load time.

Explain the affected decision, not just the pipeline

A completed refresh can sound like an all-clear to a business reader. Name the source that is late, the measure affected, and the period that remains usable. This gives colleagues a practical reporting option while preserving the visible limitation.

F Say it / transfer to a new case

A current job can use an old source

A report refreshed Friday. Sales are verified complete through Thursday, but returns only through Wednesday. Thursday returns are unknown. Both sources are complete for Wednesday.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

Friday / unknown / aligned / Wednesday

01 **Analyst:** The refresh ran on [1] _____.

02 **Manager:** Returns are complete only through [2] _____.

03 **Analyst:** Thursday returns are [3] _____.

04 **Manager:** A Wednesday-only comparison is [4] _____.

CHECK THE GAPS

1. Friday / **2.** Wednesday / **3.** unknown / **4.** aligned

Why these answers fit, page 129

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "Do we know when the missing refund delivery will arrive? I would like to tell the meeting whether the full figure will be available this afternoon." Asha: [missing response] Tomas: "Once the rows arrive, we will have business events from Tuesday loaded on a later day. We should not move those refunds into Wednesday totals." Which response belongs in the gap?

A. Not yet. The refresh timestamp is Wednesday at nine UTC, but the inputs have different coverage. Orders are verified through Tuesday; refunds are complete only through Monday.

B. Validate the backfill and document the checks. A green refresh icon should not be our only evidence that the previously incomplete reporting period is now fit for use.

C. There is no confirmed recovery estimate. I will request an update from the source owner, but I should not invent a delivery time simply because the meeting needs one.

D. Add a data-quality notice beside the affected view. State which source is delayed, its complete-through time, and which period remains usable rather than labeling the entire dashboard unreliable.

Answers and explanations, page 130

Review 1 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Metric Definitions and Business Questions" and "Data Quality and Trust". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Rules specifying exactly how a measure is calculated. Which term fits?

A. transaction-based activity

B. denominator

C. metric definition

D. internal account

2. A colleague needs the term for this meaning: Absence of unintended duplicate records. Which term fits?

A. uniqueness

B. data quality

C. coverage window

D. upstream source

3. Rebuild the exchange from "Two active-user totals, two different events". Rafael: "I have seventy-five percent on the slide: nine hundred divided by twelve hundred. Is that answering the eligible-user question, or have I changed the base?" Mei: [missing response] Rafael: "Both rates tell us something, but forty-five percent is the answer to the eligible-user question. Seventy-five percent describes the logged-in group instead." Which response belongs in the gap?

A. Assign a metric owner and record the approved version. That gives both teams a place to resolve changes before comparing totals that no longer mean the same thing.

B. That completes the reconciliation. We have preserved both correct calculations, selected the measure that answers the decision, and removed the ambiguity that created the apparent conflict.

C. That uses a different denominator. Nine hundred divided by two thousand eligible users is forty-five percent. Nine hundred divided by twelve hundred logged-in users is seventy-five percent.

D. Good. A shared reporting window removes one possible source of disagreement. The distinction here is activity type, not different month boundaries or different treatment of test accounts.

Review 1 / Meaning and context / continued

4. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "When the delivery arrives, who checks that it is complete? I do not want the warning removed just because the refresh turns green." Asha: [missing response] Tomas: "I will take the Monday-only view to the meeting, explain the Tuesday refund gap, and avoid promising a two-day net result before it is verified." Which response belongs in the gap?

A. Not yet. The refresh timestamp is Wednesday at nine UTC, but the inputs have different coverage. Orders are verified through Tuesday; refunds are complete only through Monday.

B. Validate the backfill and document the checks. A green refresh icon should not be our only evidence that the previously incomplete reporting period is now fit for use.

C. Add a data-quality notice beside the affected view. State which source is delayed, its complete-through time, and which period remains usable rather than labeling the entire dashboard unreliable.

D. Read the source cutoff. Refunds are complete through Monday at twenty-three fifty-nine UTC. The next delivery is delayed, so Tuesday is not covered by the available records.

Answers and explanations, page 161

A Read the situation

LESSON 03 / DASHBOARDS AND EXECUTIVE REPORTING

Turn an isolated monthly count into a clear performance statement with definition, target, comparison, and units.

> **Four hundred eighty needs a denominator and a target**
>
> BI analyst Jonas reviews a card labeled 480 with service director Elena. It counts September requests completed within 24 hours of receipt: 480 of 500 eligible requests. All requests have a complete observation window, and the remaining 20 finished late. The approved target is at least 95%. August had 450 timely completions among 500 eligible requests under the same definition. Jonas must report the current rate, position against target, and change from August without calling a percentage-point change a relative percentage increase.

People: Jonas, Business-intelligence analyst; Elena, Service director.

Read the record

Choose one answer per question using the facts above.

1. What is the September timely-completion rate?

A. 96%
B. 95%
C. 90%
D. 480%

2. How does September compare with the target?

A. Exactly equal to target
B. One percentage point above the 95% target
C. One percent below target
D. Six percentage points below target

3. What was the August rate?

A. 45%
B. 95%
C. 96%
D. 90%

Answers and explanations, page 131

B Vocabulary in use

CORE TERMS

KPI
Key performance indicator, a measure tied to an objective.

define a KPI

dashboard card
A compact display of a selected metric.

label the dashboard card

performance target
A specified desired level of a measure.

compare with the performance target

actual result
The observed value of the defined measure.

report the actual result

comparison period
An earlier or alternative interval used as a reference.

state the comparison period

variance
A difference from a specified reference value.

explain the target variance

percentage point
One unit of difference between percentage values.

report percentage-point changes

relative change
Change divided by the original reference value.

calculate relative change

month-on-month
Compared with the immediately preceding month.

describe month-on-month movement

year-on-year
Compared with the corresponding period a year earlier.

report year-on-year change

service-level target
An agreed performance threshold for a service measure.

track the service-level target

timely completion
Completion within the specified permitted duration.

measure timely completion

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

KPI KPI

define a KPI

Keep the meaning precise: Key performance indicator, a measure tied to an objective.

variance VARIANCE

explain the target variance

Keep the meaning precise: A difference from a specified reference value.

B Precision and professional shorthand

EXTEND YOUR RANGE

elapsed time
Duration between defined start and end events.
calculate elapsed time

observation window
The period available to observe the relevant outcome.
complete the observation window

threshold
A boundary used for a rule or decision.
apply the threshold

direction of improvement
Whether higher or lower values are favorable.
state the direction of improvement

trend
A pattern across a sequence of observations.
qualify a trend claim

seasonality
Recurring variation related to calendar patterns.
assess seasonality

run chart
A display of values ordered through time.
review the run chart

annotation
Explanatory text attached to a chart or figure.
add a relevant annotation

drill-down
Moving from a summary to more detailed data.
provide a drill-down

exception queue
Items selected for attention under a defined rule.
review the exception queue

comparability
The extent to which measures use compatible bases.
establish period comparability

executive summary
A concise account of the main evidence and decision relevance.
sharpen the executive summary

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

timely completion TIMELY COMPLETION

measure timely completion

Keep the meaning precise: Completion within the specified permitted duration.

trend TREND

qualify a trend claim

Keep the meaning precise: A pattern across a sequence of observations.

C Phrases that move the work forward

Name the measure
"This is the share completed within 24 hours."

State the start
"The clock starts when the request is received."

Give count and base
"September recorded 480 timely completions out of 500."

State the result
"The timely-completion rate is 96%."

Compare with target
"That is one percentage point above the 95% target."

Give the prior result
"August was 450 of 500, or 90%."

Use the correct change unit
"The rate rose by six percentage points."

Give the relative alternative
"That is about a 6.67% relative increase."

Protect comparability
"Both months use the same definition."

Confirm maturity
"Every request has a complete observation window."

Keep the exceptions visible
"Twenty September requests finished late."

Avoid an invented cause
"These totals do not explain the improvement."

Bound the trend
"This is a two-month comparison, not a sustained-trend claim."

Make the card interpretable
"Add the definition, denominator, target, and comparison."

Offer the next view
"The drill-down can show the late requests."

Close the readout
"September met the target and improved from August on the same basis."

Precision in real exchanges

Purpose: State the result
Useful wording: "The timely-completion rate is 96%."

Purpose: Compare with target
Useful wording: "That is one percentage point above the 95% target."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which card text is most complete?

A. 96, target unknown.

B. Up 6%, with no reference period.

C. September: 480/500 within 24 hours, 96%; target 95%; August 90%.

D. 480, excellent.

2. Which change statement is accurate?

A. Up 6% relative to August exactly

B. Up 30 percentage points because the count rose by 30

C. Up 1 percentage point from August

D. Up 6 percentage points from August

3. A colleague needs the term for this meaning: The observed value of the defined measure. Which term fits?

A. drill-down

B. threshold

C. actual result

D. month-on-month

D Choose the response that fits / continued

4. Rebuild the exchange from "Four hundred eighty needs a denominator and a target". Elena: "Our approved target is at least ninety-five percent. That means September passed, but I want the amount above the target described accurately." Jonas: [missing response] Elena: "August had four hundred fifty timely requests out of five hundred. The same definition and exclusions apply, so that earlier rate was ninety percent." Which response belongs in the gap?

A. It is one percentage point above target. Saying one percent without qualification can suggest a relative calculation, which is a different way of expressing the comparison.

B. Yes, the observation window is complete for every request here. That matters because unresolved timing can otherwise distort a rate even when the arithmetic itself is correct.

C. Preserve comparability by retaining the same start, completion rule, exclusions, and denominator. A future definition change should be flagged instead of being presented as an unexplained performance movement.

D. The KPI is timely completion: requests finished within twenty-four hours of receipt. Four hundred eighty is the timely count, not the rate or the total demand.

Answers and explanations, page 131

> **One useful distinction**
>
> September is 96%: 480/500. August is 90%: 450/500. The increase is 6 percentage points, or approximately 6.67% relative to August. September exceeds the 95% target by 1 percentage point.

E Conversation

Four hundred eighty needs a denominator and a target

Four hundred eighty needs a denominator and a target

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

KPI / drill-down / executive summary / comparison period / actual result / observation window / comparability / relative change / month-on-month / percentage point

01 **Elena:** The card says four hundred eighty with a green arrow. What does it count, and how should I interpret it?

02 **Jonas:** The [1] _____ is timely completion: requests finished within twenty-four hours of receipt. Four hundred eighty is the timely count, not the rate or the total demand.

03 **Elena:** There were five hundred eligible requests in September, and the other twenty finished late. All requests have had enough time for us to observe their outcomes.

04 **Jonas:** Then the [2] _____ is ninety-six percent, or four hundred eighty out of five hundred. We should display both the count and the base so the rate is interpretable.

05 **Elena:** Our approved target is at least ninety-five percent. That means September passed, but I want the amount above the target described accurately.

06 **Jonas:** It is one [3] _____ above target. Saying one percent without qualification can suggest a relative calculation, which is a different way of expressing the comparison.

E Conversation / continued

07 **Elena:** August had four hundred fifty timely requests out of five hundred. The same definition and exclusions apply, so that earlier rate was ninety percent.

08 **Jonas:** August is the [4] _____. September increased by six percentage points on that basis. We should name the month rather than leave the audience guessing what the arrow compares.

09 **Elena:** My draft says up six percent. Should I change that to six percentage points? I need wording I can repeat without changing the calculation.

10 **Jonas:** Correct it. The [5] _____ is approximately six point six seven percent. The absolute difference between the percentage values is six points. Both are valid when labeled, but they are not interchangeable.

Conversation continues on the next page.

Four hundred eighty needs a denominator and a target

CONVERSATION / CONTINUED

11 **Elena:** I also want to avoid the green arrow hiding the twenty late requests. Meeting the overall threshold does not mean every customer received the target service.

12 **Jonas:** We can provide a [6] _____ for those late cases. The headline can report that the target was met while the detail preserves the exceptions that still need operational attention.

13 **Elena:** Can I call it a sustained improvement, or do two months only support the month-on-month comparison? I want to recognize the result accurately.

14 **Jonas:** We can report the [7] _____ improvement. Two months do not establish a sustained trend, and these totals do not identify the cause. Recognition does not require a stronger statistical story.

15 **Elena:** Fair. We can acknowledge the result without attributing it to a particular staffing change that has not been evaluated. The definition should stay visible too.

16 **Jonas:** Preserve [8] _____ by retaining the same start, completion rule, exclusions, and denominator. A future definition change should be flagged instead of being presented as an unexplained performance movement.

17 **Elena:** The clock begins on receipt and stops on completion. We should also state that the data covers fully observed requests, not a partial month of immature outcomes.

18 **Jonas:** Yes, the [9] _____ is complete for every request here. That matters because unresolved timing can otherwise distort a rate even when the arithmetic itself is correct.

Four hundred eighty needs a denominator and a target / continued

19 **Elena:** September was ninety-six percent: one point above target, six above August, twenty requests late. The cause is unestablished.

20 **Jonas:** That is a useful [10] _____. The card will include the definition and base, and the detail will support the late-case review rather than asking the audience to infer everything from 480.

CHECK THE GAPS

1. KPI / **2.** actual result / **3.** percentage point / **4.** comparison period / **5.** relative change / **6.** drill-down / **7.** month-on-month / **8.** comparability / **9.** observation window / **10.** executive summary

Why these answers fit, page 132

F Check accuracy and scope

Four hundred eighty needs a denominator and a target

Return to the case

BI analyst Jonas reviews a card labeled 480 with service director Elena. It counts September requests completed within 24 hours of receipt: 480 of 500 eligible requests. All requests have a complete observation window, and the remaining 20 finished late. The approved target is at least 95%. August had 450 timely completions among 500 eligible requests under the same definition. Jonas must report the current rate, position against target, and change from August without calling a percentage-point change a relative percentage increase.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What is the approximate relative increase from 90% to 96%?

A. 6 percentage points expressed as 60%

B. 1%

C. 96%

D. 6.67%

2. What cannot be concluded from the supplied totals?

A. The two months used the same definition

B. The specific cause of the improvement

C. September met the target

D. Twenty September requests finished late

Answers and explanations, page 134

Listen for the distinction

Switch roles for turns 11-20. Keep the twenty late requests in the summary.

Make the interpretation portable

A useful executive card should survive being quoted outside the meeting. Include the measure and comparison in a sentence that can be repeated accurately. Avoid forcing the listener to infer whether a larger number represents improvement, demand growth, or a changed definition.

F Say it / transfer to a new case

Above target and above last month are different

A service completes 180 of 200 eligible requests on time this month. Last month was 170 of 200. The target is 88%, and both months use the same fully observed definition.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

two / five / 90 / 85

01 **Analyst:** This month is [1] _____ percent.

02 **Manager:** Last month was [2] _____ percent.

03 **Analyst:** The month-on-month increase is [3] _____ percentage points.

04 **Manager:** This month exceeds target by [4] _____ percentage points.

CHECK THE GAPS

1. 90 / **2.** 85 / **3.** five / **4.** two

Why these answers fit, page 134

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Four hundred eighty needs a denominator and a target". Elena: "Can I call it a sustained improvement, or do two months only support the month-on-month comparison? I want to recognize the result accurately." Jonas: [missing response] Elena: "Fair. We can acknowledge the result without attributing it to a particular staffing change that has not been evaluated. The definition should stay visible too." Which response belongs in the gap?

A. We can provide a drill-down for those late cases. The headline can report that the target was met while the detail preserves the exceptions that still need operational attention.

B. August is the comparison period. September increased by six percentage points on that basis. We should name the month rather than leave the audience guessing what the arrow compares.

C. Then the actual result is ninety-six percent, or four hundred eighty out of five hundred. We should display both the count and the base so the rate is interpretable.

D. We can report the month-on-month improvement. Two months do not establish a sustained trend, and these totals do not identify the cause. Recognition does not require a stronger statistical story.

Answers and explanations, page 135

A Read the situation

LESSON 04 / CAUSATION, CORRELATION, AND CAVEATS

Report a meaningful association while separating the observed difference from an unestablished causal effect.

Training attendees renewed more often

Analyst Priya reviews voluntary customer-training results with customer-success lead Marcus. Over the same renewal window, 90 of 100 eligible attendees renewed, compared with 160 of 200 eligible non-attendees. Attendance was not randomized. Baseline motivation and engagement may differ between the groups and were not measured. The records establish a 90% versus 80% renewal association, not the effect of making everyone attend. Priya recommends a reviewed evaluation design before the team uses the difference as a causal benefit claim.

People: Priya, Data analyst; Marcus, Customer-success lead.

Read the record

Choose one answer per question using the facts above.

1. What is the attendee renewal rate?

A. 80%
B. 45%
C. 10%
D. 90%

2. What is the non-attendee renewal rate?

A. 90%
B. 20%
C. 80%
D. 160%

3. Why is the causal effect not established?

A. The renewal window differs by definition.
B. Attendance was voluntary, and unmeasured baseline differences may matter.
C. The two rates are identical.
D. No customer renewed.

Answers and explanations, page 136

B Vocabulary in use

CORE TERMS

association
An observed relationship between variables or groups.

report an association

correlation
A measure or description of co-variation.

distinguish correlation from causation

causal effect
A change attributable to a specified intervention.

estimate a causal effect

confounding
Distortion from factors related to both exposure and outcome.

assess potential confounding

self-selection
Participation determined partly by people's own choices.

account for self-selection

selection bias
Systematic distortion related to how observations enter groups.

evaluate selection bias

baseline characteristic
A feature measured before the exposure or intervention.

compare baseline characteristics

motivation
A person's underlying inclination to act or engage.

assess baseline motivation

engagement
Participation or involvement under a specified definition.

define customer engagement

observational study
Analysis of naturally occurring exposures without assigned treatment.

interpret an observational study

random assignment
Allocation to groups by a defined random process.

use random assignment

treatment group
The group assigned or exposed to the studied intervention.

define the treatment group

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

association ASSOCIATION

report an association

Keep the meaning precise: An observed relationship between variables or groups.

selection bias SELECTION BIAS

evaluate selection bias

Keep the meaning precise: Systematic distortion related to how observations enter groups.

B Precision and professional shorthand

EXTEND YOUR RANGE

comparison group

The group used as a reference for the outcome.

select a comparison group

counterfactual

The outcome that would occur under an alternative exposure.

define the counterfactual question

renewal

Continuation of an eligible customer contract or relationship.

measure customer renewal

renewal window

The period over which renewal outcomes are observed.

align renewal windows

eligible customer

A customer meeting the defined inclusion conditions.

identify eligible customers

absolute difference

A difference expressed in the original measure's units.

state the absolute difference

relative lift

A difference expressed relative to a comparison value.

qualify relative lift

adjustment

Accounting for specified variables in an analysis.

assess the adjustment method

unmeasured factor

A relevant characteristic not recorded in the data.

disclose unmeasured factors

generalizability

Applicability of a result beyond the studied group.

assess generalizability

causal claim

A statement that an intervention produced an outcome.

qualify a causal claim

evaluation design

The planned method for assessing a program or intervention.

review the evaluation design

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

treatment group TREATMENT GROUP

define the treatment group

Keep the meaning precise: The group assigned or exposed to the studied intervention.

eligible customer ELIGIBLE CUSTOMER

identify eligible customers

Keep the meaning precise: A customer meeting the defined inclusion conditions.

C Phrases that move the work forward

State the observed rates
"Attendees renewed at 90%; non-attendees at 80%."

Give the counts
"That is 90 of 100 versus 160 of 200."

Name the difference
"The observed gap is ten percentage points."

Label the finding
"This is an association in the supplied data."

Identify selection
"Customers chose whether to attend."

Name the alternative explanation
"Baseline engagement may differ between the groups."

Preserve uncertainty
"Those baseline factors were not measured."

Reject the causal leap
"We cannot attribute the whole difference to training."

Avoid the opposite overclaim
"This does not prove that training has no effect."

Clarify the counterfactual
"We need to know what would have happened to comparable customers without training."

Keep the window aligned
"Both rates use the same renewal window."

Bound the audience
"The result concerns these eligible customer groups."

Qualify adjustment
"Adjusting measured factors would not automatically remove every unmeasured difference."

Recommend evaluation
"Review a design that can better isolate the training effect."

Revise the headline
"Use higher observed renewal among attendees."

Close the discussion
"Keep the association and remove the unsupported promise."

Precision in real exchanges

Purpose: Label the finding
Useful wording: "This is an association in the supplied data."

Purpose: Identify selection
Useful wording: "Customers chose whether to attend."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which headline is supported?

A. Mandatory training is proven to prevent all churn
B. Higher observed renewal among voluntary training attendees
C. Training caused a ten-point renewal increase for everyone
D. Training has no possible effect

2. What is the relative observed difference compared with 80%?

A. 12.5%
B. 10% exactly
C. 90%
D. 20 percentage points

3. A colleague needs the term for this meaning: Distortion from factors related to both exposure and outcome. Which term fits?

A. renewal
B. confounding
C. unmeasured factor
D. engagement

D Choose the response that fits / continued

4. Rebuild the exchange from "Training attendees renewed more often". Marcus: "Then the ten-point difference could reflect training, pre-existing differences, or both. We should not assume the entire gap was created by the sessions." Priya: [missing response] Marcus: "I do not want the caveat to erase the positive finding. The team has real evidence that attendees renewed more often in this group." Which response belongs in the gap?

A. Yes. The counterfactual question is what comparable customers would do without the intervention. We need a reviewed evaluation approach suited to that decision, not just a more persuasive headline.

B. Exactly. The causal effect asks what the intervention changes compared with what would otherwise have happened. This comparison does not isolate that quantity simply because the rates are clearly different.

C. It expresses relative lift against the eighty-percent reference rate. But improvement may still imply cause. Say a twelve-point-five-percent higher observed rate, and keep the nonrandom attendance qualification nearby.

D. Right. It is an unmeasured factor that may matter, not a proven explanation of every percentage point. We should avoid replacing one unsupported causal story with another equally unsupported story.

Answers and explanations, page 136

One useful distinction

The observed difference is 10 percentage points: 90%-80%. Relative to the non-attendee rate, it is 12.5%. Neither expression turns the voluntary comparison into an estimated causal benefit for mandatory training.

E Conversation

Training attendees renewed more often

Training attendees renewed more often

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

Confounding / adjustment / counterfactual / causal effect / evaluation design / relative lift / self-selection / generalizability / unmeasured factor / association

01 **Marcus:** Training attendees renewed at ninety percent, versus eighty percent for non-attendees. Can we say the sessions improved renewal by ten points?

02 **Priya:** We can report the [1] _____, but not that causal conclusion from these records. Ninety of one hundred attendees renewed; one hundred sixty of two hundred non-attendees renewed.

03 **Marcus:** Both groups had the same renewal window and eligibility rules. That makes the comparison cleaner, even though attendance was voluntary rather than assigned.

04 **Priya:** Correct, but [2] _____ remains important. Customers who chose training may have been more motivated or engaged already. Those baseline differences were not measured in the supplied data.

05 **Marcus:** Then the ten-point difference could reflect training, pre-existing differences, or both. We should not assume the entire gap was created by the sessions.

06 **Priya:** Exactly. The [3] _____ asks what the intervention changes compared with what would otherwise have happened. This comparison does not isolate that quantity simply because the rates are clearly different.

E Conversation / continued

07 **Marcus:** I do not want the caveat to erase the positive finding. The team has real evidence that attendees renewed more often in this group.

08 **Priya:** Keep that observation. [4] _____ is a reason to qualify the causal interpretation, not to deny the counts. We can state the association and the specific limitation together.

09 **Marcus:** Sales will ask whether you are saying motivation caused the whole gap. We did not measure it, so that is not our conclusion either, is it?

10 **Priya:** Right. It is an [5] _____ that may matter, not a proven explanation of every percentage point. We should avoid replacing one unsupported causal story with another equally unsupported story.

Conversation continues on the next page.

Training attendees renewed more often

CONVERSATION / CONTINUED

11 **Marcus:** For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?

12 **Priya:** It expresses [6] _____ against the eighty-percent reference rate. But improvement may still imply cause. Say a twelve-point-five-percent higher observed rate, and keep the nonrandom attendance qualification nearby.

13 **Marcus:** The business question is whether offering or requiring training would improve renewal for eligible customers. That is different from describing people who already chose to attend.

14 **Priya:** Yes. The [7] _____ question is what comparable customers would do without the intervention. We need a reviewed evaluation approach suited to that decision, not just a more persuasive headline.

15 **Marcus:** We may be able to add customer size and tenure. Would adjusting for those settle the concern, or would important differences still be missing?

16 **Priya:** A suitable [8] _____ may help with measured differences, but it does not automatically address unmeasured motivation or every source of bias. The method and assumptions need explicit review.

17 **Marcus:** And a mandatory program could reach people unlike the voluntary attendees. Even a stronger estimate in this group might not transfer unchanged to everyone.

18 **Priya:** That is a [9] _____ issue as well as a causal one. The population, intervention, and implementation matter when translating evidence into a wider operational recommendation.

Training attendees renewed more often / continued

19 **Marcus:** I will revise the slide to higher observed renewal among attendees, show both counts and rates, and recommend a better evaluation before claiming a guaranteed benefit.

20 **Priya:** Good. A reviewed [10] _____ can address the decision we actually face. We have kept the useful finding, corrected its interpretation, and made the next evidence step clear.

CHECK THE GAPS

1. association / **2.** self-selection / **3.** causal effect / **4.** Confounding / **5.** unmeasured factor / **6.** relative lift / **7.** counterfactual / **8.** adjustment / **9.** generalizability / **10.** evaluation design

Why these answers fit, page 138

F Check accuracy and scope

Training attendees renewed more often

> **Return to the case**
>
> Analyst Priya reviews voluntary customer-training results with customer-success lead Marcus. Over the same renewal window, 90 of 100 eligible attendees renewed, compared with 160 of 200 eligible non-attendees. Attendance was not randomized. Baseline motivation and engagement may differ between the groups and were not measured. The records establish a 90% versus 80% renewal association, not the effect of making everyone attend. Priya recommends a reviewed evaluation design before the team uses the difference as a causal benefit claim.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which statement about motivation is accurate?

A. It is proven to explain the entire gap.
B. It was balanced by random assignment.
C. It was measured and found identical.
D. It is an unmeasured possible source of group differences.

2. What should a revised recommendation do?

A. Preserve the association and seek a reviewed design for a causal evaluation.
B. Hide the group sizes and announce a guaranteed benefit.
C. Treat voluntary attendance as random assignment.
D. Conclude no evaluation can ever be useful.

Answers and explanations, page 139

> **Listen for the distinction**
>
> Switch roles for turns 11-20. Keep a possible confounder distinct from a proven explanation.

> **Protect the useful observation while correcting the claim**
>
> A causal caveat should not sound like all the work was pointless. State the real association first, then explain the specific competing explanation. This helps a colleague keep the finding while replacing an overconfident promise with an appropriate next evaluation.

F Say it / transfer to a new case

A difference does not supply its cause

In a voluntary program, 45 of 50 attendees renew and 80 of 100 non-attendees renew. Baseline engagement is unmeasured. Attendance was not randomized.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

ten / 80 / unestablished / 90

01 **Analyst:** Attendee renewal is [1] _____ percent.

02 **Lead:** Non-attendee renewal is [2] _____ percent.

03 **Analyst:** The observed gap is [3] _____ percentage points.

04 **Lead:** The causal effect remains [4] _____.

CHECK THE GAPS

1. 90 / **2.** 80 / **3.** ten / **4.** unestablished

Why these answers fit, page 139

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Training attendees renewed more often". Marcus: "The business question is whether offering or requiring training would improve renewal for eligible customers. That is different from describing people who already chose to attend." Priya: [missing response] Marcus: "We may be able to add customer size and tenure. Would adjusting for those settle the concern, or would important differences still be missing?" Which response belongs in the gap?

A. Yes. The counterfactual question is what comparable customers would do without the intervention. We need a reviewed evaluation approach suited to that decision, not just a more persuasive headline.

B. It expresses relative lift against the eighty-percent reference rate. But improvement may still imply cause. Say a twelve-point-five-percent higher observed rate, and keep the nonrandom attendance qualification nearby.

C. Right. It is an unmeasured factor that may matter, not a proven explanation of every percentage point. We should avoid replacing one unsupported causal story with another equally unsupported story.

D. Exactly. The causal effect asks what the intervention changes compared with what would otherwise have happened. This comparison does not isolate that quantity simply because the rates are clearly different.

Answers and explanations, page 140

Review 2 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Dashboards and Executive Reporting" and "Causation, Correlation, and Caveats". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A specified desired level of a measure. Which term fits?

A. performance target

B. annotation

C. observation window

D. relative change

2. A colleague needs the term for this meaning: A person's underlying inclination to act or engage. Which term fits?

A. association

B. relative lift

C. motivation

D. comparison group

3. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "I still need a reliable number for the meeting. Could we show Monday by itself, since both sources are verified complete through that day?" Asha: [missing response] Tomas: "The note should be near the result, not somewhere people will miss it. I will explain that Tuesday orders alone do not settle the net figure." Which response belongs in the gap?

A. Yes. That respects completeness for the shared period in this case. We can publish the Monday-only comparison and make clear that the two-day figure remains incomplete.

B. Yes. The coverage window is misaligned. We cannot interpret the result as a complete two-day net amount when the subtraction uses refunds from a shorter period.

C. Read the source cutoff. Refunds are complete through Monday at twenty-three fifty-nine UTC. The next delivery is delayed, so Tuesday is not covered by the available records.

D. Not yet. The refresh timestamp is Wednesday at nine UTC, but the inputs have different coverage. Orders are verified through Tuesday; refunds are complete only through Monday.

Review 2 / Meaning and context / continued

4. Rebuild the exchange from "Training attendees renewed more often". Marcus: "And a mandatory program could reach people unlike the voluntary attendees. Even a stronger estimate in this group might not transfer unchanged to everyone." Priya: [missing response] Marcus: "I will revise the slide to higher observed renewal among attendees, show both counts and rates, and recommend a better evaluation before claiming a guaranteed benefit." Which response belongs in the gap?

A. That is a generalizability issue as well as a causal one. The population, intervention, and implementation matter when translating evidence into a wider operational recommendation.

B. Correct, but self-selection remains important. Customers who chose training may have been more motivated or engaged already. Those baseline differences were not measured in the supplied data.

C. We can report the association, but not that causal conclusion from these records. Ninety of one hundred attendees renewed; one hundred sixty of two hundred non-attendees renewed.

D. Keep that observation. Confounding is a reason to qualify the causal interpretation, not to deny the counts. We can state the association and the specific limitation together.

Answers and explanations, page 164

A Read the situation

LESSON 05 / SQL, MODELS, AND TRANSFORMATION LOGIC

Explain row grain and join multiplicity, diagnose a duplicated aggregate, and reject a misleading distinct-value fix.

The join adds rows, not order value

Analytics engineer Lina reviews a report with analyst Owen. The complete teaching dataset has two orders: O1 is $100 and O2 is $100. Payment rows are O1: $60 and $40; O2: $100. Joining orders to payments on order ID creates three rows. Summing the repeated order amount then gives $300, although total order value is $200. SUM(DISTINCT order_amount) gives $100 because the two legitimate orders share an amount. Lina proposes aggregating payments to one row per order before joining, with key and total checks.

People: Lina, Analytics engineer; Owen, Reporting analyst.

Read the record

Choose one answer per question using the facts above.

1. How many rows result from the supplied join?

A. Two
B. Four
C. One
D. Three

2. What is the actual total order value?

A. $160
B. $200
C. $300
D. $100

3. Why does SUM(DISTINCT order_amount) give $100 here?

A. It deduplicates correctly by order ID.
B. It adds all payment amounts.
C. It restores the missing third order.
D. Both legitimate orders have the same amount, so distinct values collapse them.

Answers and explanations, page 141

B Vocabulary in use

CORE TERMS

SQL
Structured Query Language, used to query relational data.

review SQL logic

primary key
A non-null column or set of columns uniquely identifying each row.

validate the primary key

join key
The field or fields used to match tables.

choose the join key

one-to-many relationship
One entity matching multiple related records.

model a one-to-many relationship

aggregation
Combining records into summary values.

control aggregation grain

GROUP BY
SQL grouping of rows for aggregate calculations.

group by order ID

row grain
What one row represents in a dataset.

define the row grain

foreign key
A column or set of columns constrained to reference a key in a table.

check foreign-key relationships

join cardinality
The relationship in match counts between joined entities.

inspect join cardinality

fanout
Multiplication of rows caused by multiple join matches.

detect join fanout

pre-aggregation
Summarizing data before another transformation or join.

apply pre-aggregation

SUM
An aggregate that adds non-null numeric values.

check the SUM result

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

SQL SQL

review SQL logic

Keep the meaning precise: Structured Query Language, used to query relational data.

join cardinality JOIN CARDINALITY

inspect join cardinality

Keep the meaning precise: The relationship in match counts between joined entities.

B Precision and professional shorthand

EXTEND YOUR RANGE

DISTINCT

SQL removal of duplicate values or rows in its specified scope.

inspect DISTINCT semantics

COUNT

An aggregate that counts rows or non-null expressions.

distinguish COUNT variants

inner join

A join returning matching combinations from both sides.

inspect inner-join matches

left join

A join retaining left rows even without a right match.

preserve orders with a left join

null handling

Rules for processing missing or unknown values.

define null handling

filter

A condition selecting which records remain.

apply the filter

CTE

Common table expression, a named query expression.

use a CTE

data model

A representation of entities, relationships, and measures.

review the data model

fact table

A table storing business events or measurements at a defined grain.

define the fact-table grain

dimension table

A table holding descriptive attributes for analysis.

maintain dimension keys

reconciliation total

A reference total used to verify transformed output.

compare reconciliation totals

uniqueness test

A check that a defined key has no duplicates.

run a uniqueness test

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

SUM SUM

check the SUM result

Keep the meaning precise: An aggregate that adds non-null numeric values.

null handling NULL HANDLING

define null handling

Keep the meaning precise: Rules for processing missing or unknown values.

C Phrases that move the work forward

Ask about grain
"What does one row represent before the join?"

Name the key
"The tables match on order ID."

Describe the relationship
"O1 has two payment rows."

Explain the fanout
"The order row is repeated for each matching payment."

Separate data from growth
"The extra row is not an extra order."

Give the wrong total
"Summing joined order amounts produces $300."

Give the reference
"The order-level total is $200."

Reject the shortcut
"Distinct amounts do not mean distinct orders."

Show the counterexample
"Both legitimate orders have an amount of $100."

Propose the repair
"Aggregate payments to one row per order before joining."

Verify the key
"Check uniqueness of order ID in the summarized payments."

Preserve unmatched orders
"The intended order population must survive the join."

Compare totals
"Reconcile the repaired output to the source order total."

Separate measures
"Order value and payment value are different measures."

Keep null rules explicit
"Define how orders without payments are represented."

Close the review
"Fix the grain and verify the result before republishing."

Precision in real exchanges

Purpose: Explain the fanout
Useful wording: "The order row is repeated for each matching payment."

Purpose: Separate data from growth
Useful wording: "The extra row is not an extra order."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which operation explains the $300 result?

A. Summing the actual 60, 40, and 100 payment amounts
B. Adding the two original order amounts once each
C. Counting distinct order IDs
D. Summing order amounts after O1 is repeated for two payments

2. Why is summing distinct amounts not a valid repair?

A. It changes O2 into a zero-value order.
B. It merges different orders that happen to have equal values.
C. It always preserves every order identity.
D. It forces payment rows to become unique by ID.

3. A colleague needs the term for this meaning: A column or set of columns constrained to reference a key in a table. Which term fits?

A. fact table
B. aggregation
C. foreign key
D. inner join

D Choose the response that fits / continued

4. Rebuild the exchange from "The join adds rows, not order value". Owen: "I summed the order amount after the join. Each resulting row displays one hundred, so the three displayed amounts add to three hundred." Lina: [missing response] Owen: "SUM DISTINCT brought it down to one hundred. That looks worse: we have two separate orders worth one hundred each. Have I removed the wrong duplicate?" Which response belongs in the gap?

A. That is join fanout. The payment rows are legitimate, but the order-level measure is repeated. We need to preserve its meaning instead of treating each joined occurrence as new value.

B. Use the original reconciliation total as a reference and inspect differences. A plausible-looking grand total alone is not enough if missing orders and duplicated orders happen to offset each other.

C. Then choose and test the appropriate left join. It can preserve unmatched orders, but it would not by itself solve multiple matches. Population preservation and multiplicity are separate checks.

D. Keep the aggregation tied to the intended entity. An order-value report and a payment-value report answer different questions, even when their totals happen to match in this example.

Answers and explanations, page 141

One useful distinction

The joined order amounts are 100, 100, and 100: O1 appears twice because it has two payments. The correct order total is 200. The payment amounts themselves are 60, 40, and 100, also totaling 200 for this dataset.

E Conversation

The join adds rows, not order value

The join adds rows, not order value

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

row grain / aggregation / reconciliation total / fanout / DISTINCT / one-to-many relationship / pre-aggregation / join key / uniqueness test / left join

01 **Owen:** The report jumped from two hundred dollars of order value to three hundred after I added payment details. I thought the join would only add columns.

02 **Lina:** Check the [1] _____. Before the join, each row represents one order. After matching O1 to two payment records, that order appears in two joined rows.

03 **Owen:** O1 is one hundred dollars with payments of sixty and forty. O2 is another one-hundred-dollar order with a single one-hundred-dollar payment.

04 **Lina:** That is a [2] _____. The join produces three rows: two for O1 and one for O2. The third row is not evidence of another business order.

05 **Owen:** I summed the order amount after the join. Each resulting row displays one hundred, so the three displayed amounts add to three hundred.

06 **Lina:** That is join [3] _____. The payment rows are legitimate, but the order-level measure is repeated. We need to preserve its meaning instead of treating each joined occurrence as new value.

E Conversation / continued

07 **Owen:** SUM DISTINCT brought it down to one hundred. That looks worse: we have two separate orders worth one hundred each. Have I removed the wrong duplicate?

08 **Lina:** Exactly. [4] _____ applies to the selected value here. Two different orders can both be worth one hundred dollars, so amount equality does not establish duplicate business identity.

09 **Owen:** The correct order total is two hundred. The payment amounts also total two hundred in this small dataset, but that does not make the measures identical.

10 **Lina:** Keep the [5] _____ tied to the intended entity. An order-value report and a payment-value report answer different questions, even when their totals happen to match in this example.

Conversation continues on the next page.

The join adds rows, not order value

CONVERSATION / CONTINUED

11 **Owen:** For the order report, I can summarize the payment side first so there is one payment-summary row for each order ID.

12 **Lina:** Yes, use [6] _____. Then verify that the payment summary has only one row per order. Joining two aligned grains avoids repeating the original order amount for every payment.

13 **Owen:** The grouping field should be order ID, not amount. Grouping by amount would combine the two orders again because they both have the same value.

14 **Lina:** Correct. The [7] _____ must preserve the intended relationship. Test the summary key and check that the matching logic uses the appropriate identifier on both sides.

15 **Owen:** Both orders in this extract have payments. But the full report must still include an order without a payment if it belongs in the reporting population.

16 **Lina:** Then choose and test the appropriate [8] _____. It can preserve unmatched orders, but it would not by itself solve multiple matches. Population preservation and multiplicity are separate checks.

17 **Owen:** I will check both the two-order count and the two-hundred-dollar total. Could those totals look right even if one missing order offsets a duplicate?

18 **Lina:** Use the original [9] _____ as a reference and inspect differences. A plausible-looking grand total alone is not enough if missing orders and duplicated orders happen to offset each other.

19 **Owen:** My explanation will say the join repeated an order-level amount; it did not create revenue. I will withhold the inflated result while the transformation is corrected.

20 **Lina:** Add a [10] _____ for the summarized order key and document the grain. That makes the repair repeatable rather than relying on someone noticing another surprising total later.

CHECK THE GAPS

1. row grain / **2.** one-to-many relationship / **3.** fanout / **4.** DISTINCT / **5.** aggregation / **6.** pre-aggregation / **7.** join key / **8.** left join / **9.** reconciliation total / **10.** uniqueness test

Why these answers fit, page 143

F Check accuracy and scope

The join adds rows, not order value

Return to the case

Analytics engineer Lina reviews a report with analyst Owen. The complete teaching dataset has two orders: O1 is $100 and O2 is $100. Payment rows are O1: $60 and $40; O2: $100. Joining orders to payments on order ID creates three rows. Summing the repeated order amount then gives $300, although total order value is $200. SUM(DISTINCT order_amount) gives $100 because the two legitimate orders share an amount. Lina proposes aggregating payments to one row per order before joining, with key and total checks.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which proposed repair fits this report?

A. Replace the inner join with a left join and sum the repeated order values.
B. Summarize payments by order ID, then join at one row per order and verify totals.
C. Keep the payment-detail join and sum distinct order amounts.
D. Group payments by amount before joining to each order amount.

2. Which statement about a left join is accurate?

A. It can preserve unmatched orders but still duplicate an order with multiple matches.
B. It always guarantees one row per order.
C. It prevents every duplicate amount.
D. It discards every left-side order without payment.

Answers and explanations, page 144

Listen for the distinction

Switch roles for turns 11-20. Stress one row per order and both reconciliation checks.

Explain the row, then the total

When a report suddenly grows after a join, avoid treating the larger figure as business growth or blaming a source team without checking the transformation. A small concrete example lets technical and business colleagues inspect the same logic and agree on a verifiable repair.

F Say it / transfer to a new case

Equal amounts can belong to different orders

Order A is $80 with two $40 payments. Order B is $80 with one $80 payment. An order-to-payment join yields three rows. The report must count each order amount once.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

160 / 80 / 240 / identity

01 **Engineer:** The correct order total is $[1] _____.

02 **Analyst:** The joined sum of order amounts is $[2] _____.

03 **Engineer:** The sum of distinct amount values is $[3] _____.

04 **Analyst:** The repair must preserve order [4] _____.

CHECK THE GAPS

1. 160 / **2.** 240 / **3.** 80 / **4.** identity

Why these answers fit, page 144

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "The join adds rows, not order value". Owen: "The grouping field should be order ID, not amount. Grouping by amount would combine the two orders again because they both have the same value." Lina: [missing response] Owen: "Both orders in this extract have payments. But the full report must still include an order without a payment if it belongs in the reporting population." Which response belongs in the gap?

A. Correct. The join key must preserve the intended relationship. Test the summary key and check that the matching logic uses the appropriate identifier on both sides.

B. That is a one-to-many relationship. The join produces three rows: two for O1 and one for O2. The third row is not evidence of another business order.

C. Exactly. DISTINCT applies to the selected value here. Two different orders can both be worth one hundred dollars, so amount equality does not establish duplicate business identity.

D. Check the row grain. Before the join, each row represents one order. After matching O1 to two payment records, that order appears in two joined rows.

Answers and explanations, page 145

A Read the situation

LESSON 06 / EXPERIMENT READOUTS AND STATISTICAL THINKING

Present an experiment estimate with its interval, practical threshold, and pre-agreed decision rule.

A positive estimate is not the rollout rule

Experiment analyst Theo reviews a completed randomized test with product director Hana. The validated readout estimates a 1.2-percentage-point increase in completion, with a 95% confidence interval from -0.4 to +2.8 points. The pre-agreed rollout rule requires the interval lower bound to exceed +1.0 point, with guardrails passing. Guardrails passed, but the lower-bound condition did not. The slide currently shows only +1.2. The test has reached its planned endpoint; no unplanned extension or new rollout approval is authorized.

People: Theo, Experiment analyst; Hana, Product director.

Read the record

Choose one answer per question using the facts above.

1. What is the point estimate?

A. -0.4 percentage points
B. +1.0 percentage point
C. +1.2 percentage points
D. +2.8 percentage points

2. Did the lower-bound rule pass?

A. No; -0.4 does not exceed +1.0.
B. Yes; the point estimate exceeds +1.0.
C. Yes; every positive estimate passes.
D. No; guardrails failed.

3. What stage has the test reached?

A. A completed rollout
B. Its planned endpoint
C. An unplanned first-day peek
D. An automatically approved extension

Answers and explanations, page 146

B Vocabulary in use

CORE TERMS

randomized experiment
A study allocating exposure through a random process.

interpret a randomized experiment

point estimate
A single value estimating an unknown quantity.

report the point estimate

interval estimate
A range estimating an unknown quantity under a method.

report an interval estimate

confidence interval
An interval from a method with specified long-run coverage.

interpret a confidence interval

confidence level
The nominal long-run coverage rate of the interval method.

state the confidence level

lower bound
The smaller endpoint of an interval.

compare the lower bound

upper bound
The larger endpoint of an interval.

report the upper bound

effect size
The magnitude of a difference or relationship.

assess effect size

no-change value
The parameter value representing no effect on the stated scale.

identify the no-change value

practical threshold
The effect boundary relevant to a business decision.

set a practical threshold

decision rule
A predefined condition governing a conclusion or action.

apply the decision rule

guardrail
A measure limiting unacceptable side effects.

check the guardrails

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

randomized experiment RANDOMIZED EXPERIMENT

interpret a randomized experiment

Keep the meaning precise: A study allocating exposure through a random process.

lower bound LOWER BOUND

compare the lower bound

Keep the meaning precise: The smaller endpoint of an interval.

B Precision and professional shorthand

EXTEND YOUR RANGE

primary outcome

The main measure used to assess the intervention.

specify the primary outcome

planned endpoint

The pre-agreed point for completing the analysis or test.

respect the planned endpoint

statistical significance

A result meeting a specified statistical testing criterion.

qualify statistical significance

practical significance

Importance of an effect for a real decision.

assess practical significance

null hypothesis

A specified claim assessed by a statistical test.

state the null hypothesis

standard error

The standard deviation of an estimator's sampling distribution, usually estimated.

report the standard error

sampling variability

Variation in estimates across possible samples.

account for sampling variability

statistical power

Probability of detecting a specified effect under assumptions.

plan statistical power

minimum detectable effect

An effect size used in the study's power planning.

specify the minimum detectable effect

sequential analysis

A method allowing planned repeated statistical decisions.

use a valid sequential analysis

equivalence

Similarity within a predefined acceptable difference margin.

test equivalence appropriately

inconclusive result

Evidence insufficient for the specified conclusion.

explain an inconclusive result

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

guardrail GUARDRAIL

check the guardrails

Keep the meaning precise: A measure limiting unacceptable side effects.

null hypothesis NULL HYPOTHESIS

state the null hypothesis

Keep the meaning precise: A specified claim assessed by a statistical test.

C Phrases that move the work forward

Report the center
"The estimated effect is plus 1.2 percentage points."

Show the uncertainty
"The 95% interval runs from minus 0.4 to plus 2.8 points."

Name the no-change value
"The interval includes zero."

Keep the scale explicit
"These are percentage points, not relative percentages."

State the actual rule
"The lower bound must exceed plus 1.0 point."

Apply the rule
"The lower-bound condition was not met."

Preserve the guardrail result
"The guardrails passed."

Separate the conditions
"Passing guardrails does not replace the effect requirement."

Avoid declaring a winner
"This readout does not authorize rollout under our rule."

Avoid declaring no effect
"It also does not prove the true effect is zero."

Respect the endpoint
"The planned test has ended."

Reject opportunistic extension
"We cannot keep collecting until the result looks favorable."

Discuss next work honestly
"Any new study needs an explicitly reviewed design."

Keep the interval visible
"Place both bounds beside the estimate."

Distinguish decision from interest
"The signal may merit follow-up without passing this launch rule."

Close the readout
"Report the estimate, interval, guardrails, and unmet decision condition together."

Precision in real exchanges

Purpose: Keep the scale explicit
Useful wording: "These are percentage points, not relative percentages."

Purpose: State the actual rule
Useful wording: "The lower bound must exceed plus 1.0 point."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which slide is most faithful to the readout?

A. +1.2 points; 95% interval -0.4 to +2.8; rollout approved because guardrails pass

B. +1.2% relative lift; 95% interval -0.4 to +2.8 points; criterion unmet

C. +1.2 points; 95% interval +0.4 to +2.8; guardrails pass; criterion unmet

D. +1.2 points; 95% interval -0.4 to +2.8; guardrails pass; rollout criterion unmet

2. Why does the +1.2 estimate not satisfy the rule?

A. Every positive effect must be ignored.

B. Guardrails always determine effect size.

C. The rule uses the interval lower bound, which is -0.4.

D. The rule requires the upper bound to be negative.

3. A colleague needs the term for this meaning: An interval from a method with specified long-run coverage. Which term fits?

A. confidence interval

B. minimum detectable effect

C. no-change value

D. statistical significance

D Choose the response that fits / continued

4. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "Remind me of the actual rollout rule. I remember that we set a one-percentage-point threshold and also required the guardrails to pass." Theo: [missing response] Hana: "The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement." Which response belongs in the gap?

A. Correct. The confidence interval belongs beside the estimate. It prevents us from presenting a promising central value as if the uncertainty were negligible or entirely favorable.

B. That preserves practical significance as a separate question. The signal may merit well-designed follow-up, but this completed test has not passed the agreed rule for rollout.

C. The lower bound must exceed plus one point zero, with guardrails passing. Our lower endpoint is minus zero point four, so that condition was not met.

D. Good. The practical threshold should be displayed on the same scale. Statistical uncertainty and business value are related in the decision, but they are not interchangeable labels.

Answers and explanations, page 146

> **One useful distinction**
>
> The point estimate is +1.2 points, but the interval runs from -0.4 to +2.8. It includes zero and values below the practical threshold. Under the supplied rule, a passing guardrail cannot compensate for a failing lower bound.

E Conversation

A positive estimate is not the rollout rule

A positive estimate is not the rollout rule

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

lower bound / practical significance / decision rule / planned endpoint / practical threshold / sequential analysis / no-change value / confidence interval / inconclusive result / point estimate

01 **Hana:** The slide says plus one point two, which looks promising. Can we call the test a win and move to rollout now that it has finished?

02 **Theo:** Show the [1] _____ with the interval. The estimated increase is one point two percentage points, but the ninety-five-percent interval extends from minus zero point four to plus two point eight.

03 **Hana:** That range includes a small negative effect and zero as well as useful positive outcomes. The current slide hides those possibilities by showing only the center.

04 **Theo:** Correct. The [2] _____ belongs beside the estimate. It prevents us from presenting a promising central value as if the uncertainty were negligible or entirely favorable.

05 **Hana:** Remind me of the actual rollout rule. I remember that we set a one-percentage-point threshold and also required the guardrails to pass.

06 **Theo:** The [3] _____ must exceed plus one point zero, with guardrails passing. Our lower endpoint is minus zero point four, so that condition was not met.

E Conversation / continued

07 **Hana:** The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement.

08 **Theo:** Exactly. Apply the whole [4] _____. We can acknowledge the guardrail result while saying the test did not authorize rollout under the standard agreed before seeing these results.

09 **Hana:** Then should the slide say no effect? Zero is inside the interval, but so are positive effects. I do not want to replace one overclaim with another.

10 **Theo:** No. The [5] _____ is included, but useful positive effects are included too. The readout does not establish exact zero or show that the two experiences are equivalent.

Conversation continues on the next page.

A positive estimate is not the rollout rule

CONVERSATION / CONTINUED

11 **Hana:** Then the decision is clearer than the underlying effect. We know our agreed rollout rule failed, while the magnitude and even direction remain uncertain.

12 **Theo:** That is an [6] _____ for the desired benefit claim, not permission to hide the estimate. We should report what the evidence supports and what decision it does not support.

13 **Hana:** Could we extend it until the lower bound clears one point? Or would that change the rule after we have already seen the result?

14 **Theo:** We reached the [7] _____. Continuing until the result looks favorable changes the statistical behavior. There is no authorized extension or approved repeated-decision method in the supplied plan.

15 **Hana:** A new study might still be justified, but it would need a clear design and rationale rather than an attempt to rescue the original headline.

16 **Theo:** Yes. An appropriate [8] _____ can support planned repeated decisions, but that is not what was specified here. Any new evaluation needs its own reviewed assumptions and rules.

17 **Hana:** I will also keep the units visible. Plus one point two means percentage points, not a one-point-two-percent relative increase from an unstated baseline.

18 **Theo:** Good. The [9] _____ should be displayed on the same scale. Statistical uncertainty and business value are related in the decision, but they are not interchangeable labels.

A positive estimate is not the rollout rule / continued

19 **Hana:** My revised readout will show the center, both bounds, passing guardrails, and the unmet rollout condition. No launch approval or no-effect conclusion will be implied.

20 **Theo:** That preserves [10] _____ as a separate question. The signal may merit well-designed follow-up, but this completed test has not passed the agreed rule for rollout.

CHECK THE GAPS

1. point estimate / **2.** confidence interval / **3.** lower bound / **4.** decision rule / **5.** no-change value / **6.** inconclusive result / **7.** planned endpoint / **8.** sequential analysis / **9.** practical threshold / **10.** practical significance

Why these answers fit, page 147

F Check accuracy and scope

A positive estimate is not the rollout rule

Return to the case

Experiment analyst Theo reviews a completed randomized test with product director Hana. The validated readout estimates a 1.2-percentage-point increase in completion, with a 95% confidence interval from -0.4 to +2.8 points. The pre-agreed rollout rule requires the interval lower bound to exceed +1.0 point, with guardrails passing. Guardrails passed, but the lower-bound condition did not. The slide currently shows only +1.2. The test has reached its planned endpoint; no unplanned extension or new rollout approval is authorized.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What would be an unsupported next step?

A. Report that the decision condition was unmet.

B. Review a properly designed follow-up study.

C. Keep the interval visible on the slide.

D. Extend the finished test repeatedly until the original method produces a favorable result.

2. Which statement about zero is correct?

A. Inclusion proves the treatment and control are identical.

B. Inclusion means no further study can be useful.

C. Inclusion makes the interval a probability distribution over customers.

D. Its inclusion does not prove either exact zero effect or equivalence.

Answers and explanations, page 149

Listen for the distinction

Switch roles for turns 11-20. Contrast an unmet rollout rule with proof of no effect.

Distinguish a promising signal from a passed rule

A positive estimate can be worth discussing without being a win under the actual decision standard. Show the interval and threshold together. Correcting an overconfident slide should preserve the evidence, not replace optimism with an equally unsupported claim of no effect.

F Say it / transfer to a new case

Read the endpoint named in the rule

An experiment estimates +0.8 points with a 95% interval from -0.2 to +1.8. The rule requires the lower bound to exceed +0.5, and guardrails pass. The planned test has ended.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

0.8 / 0.5 / unmet / 0.2

01 **Analyst:** The point estimate is plus [1] _____ points.

02 **Director:** The lower bound is minus [2] _____ points.

03 **Analyst:** The required boundary is plus [3] _____ points.

04 **Director:** The rollout condition is [4] _____.

CHECK THE GAPS

1. 0.8 / **2.** 0.2 / **3.** 0.5 / **4.** unmet

Why these answers fit, page 149

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "Could we extend it until the lower bound clears one point? Or would that change the rule after we have already seen the result?" Theo: [missing response] Hana: "A new study might still be justified, but it would need a clear design and rationale rather than an attempt to rescue the original headline." Which response belongs in the gap?

A. No. The no-change value is included, but useful positive effects are included too. The readout does not establish exact zero or show that the two experiences are equivalent.

B. We reached the planned endpoint. Continuing until the result looks favorable changes the statistical behavior. There is no authorized extension or approved repeated-decision method in the supplied plan.

C. Exactly. Apply the whole decision rule. We can acknowledge the guardrail result while saying the test did not authorize rollout under the standard agreed before seeing these results.

D. Yes. An appropriate sequential analysis can support planned repeated decisions, but that is not what was specified here. Any new evaluation needs its own reviewed assumptions and rules.

Answers and explanations, page 150

Review 3 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "SQL, Models, and Transformation Logic" and "Experiment Readouts and Statistical Thinking". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A non-null column or set of columns uniquely identifying each row. Which term fits?

A. primary key
B. COUNT
C. fanout
D. data model

2. A colleague needs the term for this meaning: The magnitude of a difference or relationship. Which term fits?

A. primary outcome
B. randomized experiment
C. sampling variability
D. effect size

3. Rebuild the exchange from "Training attendees renewed more often". Marcus: "Sales will ask whether you are saying motivation caused the whole gap. We did not measure it, so that is not our conclusion either, is it?" Priya: [missing response] Marcus: "For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?" Which response belongs in the gap?

A. It expresses relative lift against the eighty-percent reference rate. But improvement may still imply cause. Say a twelve-point-five-percent higher observed rate, and keep the nonrandom attendance qualification nearby.
B. Yes. The counterfactual question is what comparable customers would do without the intervention. We need a reviewed evaluation approach suited to that decision, not just a more persuasive headline.
C. Exactly. The causal effect asks what the intervention changes compared with what would otherwise have happened. This comparison does not isolate that quantity simply because the rates are clearly different.
D. Right. It is an unmeasured factor that may matter, not a proven explanation of every percentage point. We should avoid replacing one unsupported causal story with another equally unsupported story.

Review 3 / Meaning and context / continued

4. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "I will also keep the units visible. Plus one point two means percentage points, not a one-point-two-percent relative increase from an unstated baseline." Theo: [missing response] Hana: "My revised readout will show the center, both bounds, passing guardrails, and the unmet rollout condition. No launch approval or no-effect conclusion will be implied." Which response belongs in the gap?

A. Good. The practical threshold should be displayed on the same scale. Statistical uncertainty and business value are related in the decision, but they are not interchangeable labels.

B. The lower bound must exceed plus one point zero, with guardrails passing. Our lower endpoint is minus zero point four, so that condition was not met.

C. Correct. The confidence interval belongs beside the estimate. It prevents us from presenting a promising central value as if the uncertainty were negligible or entirely favorable.

D. Exactly. Apply the whole decision rule. We can acknowledge the guardrail result while saying the test did not authorize rollout under the standard agreed before seeing these results.

Answers and explanations, page 167

A Read the situation

LESSON 07 / DATA GOVERNANCE AND PRIVACY

Specify a purpose-limited data request, distinguish pseudonymization from anonymity, and preserve approval and retention controls.

> **The analysis needs weekly counts, not customer names**
>
> Analyst Sara asks data steward Malik for a full customer export to compare weekly late-resolution rates by product tier. The supplied analysis can use week, tier, eligible-case count, and late-case count; names, emails, phone numbers, and free-text comments are unnecessary. A team proposal uses a reviewed aggregate table in the approved workspace with 30-day access, subject to owner approval and disclosure review. Approval is pending. Small groups still require review. Replacing names with stable IDs would not automatically make row-level data anonymous or remove controls.

People: Sara, Data analyst; Malik, Data steward.

Read the record

Choose one answer per question using the facts above.

1. Which fields directly support the supplied analysis?

A. Passwords and private payment credentials
B. Week, tier, eligible-case count, and late-case count
C. Every customer's name and phone number
D. Full free-text comments with unrelated details

2. What is the current approval status?

A. Pending
B. Automatic because the data is aggregated
C. Granted by the analyst's request
D. Permanent approval for every future purpose

3. What does replacing names with stable IDs establish?

A. Every row can be publicly released.
B. It does not automatically establish anonymity.
C. All access controls become unnecessary.
D. Re-identification is impossible in every context.

Answers and explanations, page 151

B Vocabulary in use

CORE TERMS

data governance
Responsibilities and rules for managing data use.

establish data governance

data steward
A role supporting defined data quality and handling responsibilities.

consult the data steward

data owner
The role accountable for specified data decisions.

obtain data-owner approval

purpose limitation
Restricting data use to defined appropriate purposes.

document the analysis purpose

data minimization
Limiting data to what the stated purpose needs.

apply data minimization

personal data
Information relating to an identifiable person under the applicable rules.

assess personal-data handling

direct identifier
A field that directly identifies a person.

remove unnecessary direct identifiers

indirect identifier
A field that may identify someone when combined with other information.

review indirect identifiers

pseudonymization
Replacing identifying details while retaining possible linkage.

assess pseudonymization safeguards

anonymization
Processing intended to prevent identification under the relevant standard.

evaluate anonymization

re-identification risk
The possibility of connecting data back to an individual.

assess re-identification risk

aggregation
Combining records into group-level summaries.

review aggregation level

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

data governance DATA GOVERNANCE

establish data governance

Keep the meaning precise: Responsibilities and rules for managing data use.

personal data PERSONAL DATA

assess personal-data handling

Keep the meaning precise: Information relating to an identifiable person under the applicable rules.

B Precision and professional shorthand

EXTEND YOUR RANGE

small cell

A reported group with few contributing observations.

review small cells

disclosure review

Assessment of risks from releasing or sharing information.

conduct disclosure review

access scope

The specific data and actions a permission covers.

limit access scope

retention period

The duration for which data is kept under defined rules.

document the retention period

access expiry

The point when an access permission ends.

enforce access expiry

approved workspace

An environment authorized for the stated data work.

use the approved workspace

data lineage

The traceable path of data through sources and transformations.

document data lineage

data catalog

An organized inventory of datasets and their metadata.

consult the data catalog

processing purpose

The defined reason for collecting or using data.

state the processing purpose

permitted audience

The people authorized to receive a result or dataset.

define the permitted audience

export restriction

A limit on moving data outside an approved environment.

enforce export restrictions

secondary use

A use beyond the original stated purpose.

review proposed secondary use

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

aggregation AGGREGATION

review aggregation level

Keep the meaning precise: Combining records into group-level summaries.

access expiry ACCESS EXPIRY

enforce access expiry

Keep the meaning precise: The point when an access permission ends.

C Phrases that move the work forward

Define the purpose
"We are comparing weekly late-resolution rates by tier."

Limit identifiers
"Names and contact details are unnecessary for this question."

Preserve the denominator
"Include eligible-case counts as well as late-case counts."

State the environment
"The work belongs in the approved workspace."

Separate duration rules
"Access expiry is not automatically the data-retention rule."

Avoid an anonymity claim
"Stable IDs do not automatically make the records anonymous."

Keep exports governed
"Do not assume approval permits an external export."

Record the source
"Keep the source and transformation lineage."

Link fields to the task
"The calculation needs counts by week and tier."

Offer an alternative
"Use the reviewed aggregate table."

Keep approval visible
"Owner approval is still pending."

Bound access
"The proposed access duration is thirty days."

Check group disclosure
"Small groups still need disclosure review."

Control the audience
"Confirm who may receive the output."

Address new purposes
"A different question may need a new review."

Close the request
"Submit the purpose, four fields, workspace, duration, and review conditions."

Precision in real exchanges

Purpose: Offer an alternative
Useful wording: "Use the reviewed aggregate table."

Purpose: Preserve the denominator
Useful wording: "Include eligible-case counts as well as late-case counts."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which request best matches the supplied need?

A. Reviewed weekly tier-level counts in the approved workspace, with pending approvals and 30-day access

B. A full customer export with unlimited future use

C. Names and phone numbers without case counts

D. Public release of stable-ID records without review

2. Why include eligible-case count?

A. It provides the denominator for each rate.

B. It identifies every customer's phone number.

C. It proves all small groups are anonymous.

D. It replaces owner approval.

3. A colleague needs the term for this meaning: Restricting data use to defined appropriate purposes. Which term fits?

A. access scope

B. processing purpose

C. pseudonymization

D. purpose limitation

D Choose the response that fits / continued

4. Rebuild the exchange from "The analysis needs weekly counts, not customer names". Sara: "An aggregate table would work for this comparison. I do not need individual rows if the counts use the agreed definitions and can be traced to their source." Malik: [missing response] Sara: "So a small tier in one week should not automatically be published just because its row contains counts rather than names. The audience and context still matter." Which response belongs in the gap?

A. Then data minimization supports a narrower request. We should retain what makes the analysis valid, including its denominator, without adding unrelated personal fields simply because they might be interesting later.

B. Correct. A disclosure review remains part of the proposal. Do not treat a change in table shape as an automatic conclusion that all privacy and handling conditions have disappeared.

C. That aggregation is a useful alternative, subject to review. We still need to check the output, especially where a group is small enough to raise identification or disclosure concerns.

D. Define the permitted audience and the output route. The conditions should remain attached to the analysis even after the calculations are complete and somebody wants to reuse a chart.

Answers and explanations, page 151

> **One useful distinction**
>
> The rate needs late-case count divided by eligible-case count within each week and tier. Names and contact details do not support that supplied calculation. Data minimization means enough relevant data for the purpose, not an unusably stripped dataset.

E Conversation

The analysis needs weekly counts, not customer names

The analysis needs weekly counts, not customer names

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

access scope / Access expiry / disclosure review / pseudonymization / data lineage / aggregation / secondary use / permitted audience / processing purpose / data minimization

01 **Sara:** I requested the full customer export for weekly late-resolution rates by product tier. I thought extra fields would give us flexibility.

02 **Malik:** Start with the [1] _____. The supplied question needs week, tier, eligible-case count, and late-case count. Which part requires names, contact details, or unrestricted free-text comments?

03 **Sara:** Those extra fields do not enter the rate. I need late cases divided by eligible cases within each week and tier.

04 **Malik:** Then [2] _____ supports a narrower request. We should retain what makes the analysis valid, including its denominator, without adding unrelated personal fields simply because they might be interesting later.

05 **Sara:** An aggregate table would work for this comparison. I do not need individual rows if the counts use the agreed definitions and can be traced to their source.

06 **Malik:** That [3] _____ is a useful alternative, subject to review. We still need to check the output, especially where a group is small enough to raise identification or disclosure concerns.

E Conversation / continued

07 **Sara:** So a small tier in one week should not automatically be published just because its row contains counts rather than names. The audience and context still matter.

08 **Malik:** Correct. A [4] _____ remains part of the proposal. Do not treat a change in table shape as an automatic conclusion that all privacy and handling conditions have disappeared.

09 **Sara:** If someone later asks for individual records, could replacing customer names with stable IDs make that request anonymous and remove the approval requirement?

10 **Malik:** That is usually a [5] _____ question, not an automatic anonymity claim. Stable identifiers can still support linkage, and the applicable assessment and access controls must remain explicit.

Conversation continues on the next page.

The analysis needs weekly counts, not customer names

CONVERSATION / CONTINUED

11 **Sara:** For this task, I will stay with the reviewed aggregate fields. The proposal also specifies the approved workspace rather than an unrestricted download to any location.

12 **Malik:** Keep that in the [6] _____. Owner approval is still pending. A technically convenient format does not authorize an export or make the request approved before the decision is recorded.

13 **Sara:** Does thirty-day access also mean deleting the data on day thirty? I need to distinguish the permission expiry from the actual retention rule.

14 **Malik:** Exactly. [7] _____ and data retention are separate questions. The actual retention and deletion arrangements need their own approved basis rather than being inferred from this access duration.

15 **Sara:** I will also say who needs the results. The service manager needs the reviewed weekly comparison, but that does not imply that anyone may receive the underlying data.

16 **Malik:** Define the [8] _____ and the output route. The conditions should remain attached to the analysis even after the calculations are complete and somebody wants to reuse a chart.

17 **Sara:** If we later need customer-level linkage, I will bring a new purpose and field list. This request should not become permission for unrelated analysis.

18 **Malik:** That would be a [9] _____ review. We can assess the new need on its facts, but the current weekly-rate purpose should not become a blanket authorization for every future analysis.

The analysis needs weekly counts, not customer names / continued

19 **Sara:** My request will list the four fields, rate, workspace, thirty-day access, audience, and required reviews. Approval remains pending.

20 **Malik:** Include the [10] _____ for the aggregate counts as well. That preserves both analytical trust and the handling boundary while giving you the information the actual question needs.

CHECK THE GAPS

1. processing purpose / **2.** data minimization / **3.** aggregation / **4.** disclosure review / **5.** pseudonymization / **6.** access scope / **7.** Access expiry / **8.** permitted audience / **9.** secondary use / **10.** data lineage

Why these answers fit, page 153

F Check accuracy and scope

The analysis needs weekly counts, not customer names

Return to the case

Analyst Sara asks data steward Malik for a full customer export to compare weekly late-resolution rates by product tier. The supplied analysis can use week, tier, eligible-case count, and late-case count; names, emails, phone numbers, and free-text comments are unnecessary. A team proposal uses a reviewed aggregate table in the approved workspace with 30-day access, subject to owner approval and disclosure review. Approval is pending. Small groups still require review. Replacing names with stable IDs would not automatically make row-level data anonymous or remove controls.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which statement about aggregation is accurate?

A. It can reduce detail but still needs the stated disclosure review.
B. Every aggregate is automatically safe to publish.
C. It always preserves every individual identifier.
D. It removes the need for an analysis purpose.

2. What does 30-day access specify?

A. Permanent permission after the first month
B. Approval already granted to all employees
C. The proposed duration of permission in this case
D. A universal legal deletion deadline

Answers and explanations, page 154

Listen for the distinction

Switch roles for turns 11-20. Keep access expiry distinct from retention.

Tie each field to the question

A constructive data-access discussion explains what information the analysis requires and offers a usable alternative. Avoid assuming that analysts need everything or that every aggregate is harmless. Keep the purpose, approval, audience, and handling conditions visible in the same request.

F Say it / transfer to a new case

A denominator is necessary; a phone number is not

An analyst needs monthly complaint rates by region. The approved design uses month, region, complaint count, and eligible-order count. Names and phone numbers are unnecessary. Access approval is pending; proposed access lasts 14 days.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

pending / 14 / count / unnecessary

01 **Analyst:** The rate denominator is the eligible-order [1] _____.

02 **Steward:** Phone numbers are [2] _____ for this question.

03 **Analyst:** The proposed access period is [3] _____ days.

04 **Steward:** Approval remains [4] _____.

CHECK THE GAPS

1. count / **2.** unnecessary / **3.** 14 / **4.** pending

Why these answers fit, page 154

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "The analysis needs weekly counts, not customer names". Sara: "Does thirty-day access also mean deleting the data on day thirty? I need to distinguish the permission expiry from the actual retention rule." Malik: [missing response] Sara: "I will also say who needs the results. The service manager needs the reviewed weekly comparison, but that does not imply that anyone may receive the underlying data." Which response belongs in the gap?

A. That aggregation is a useful alternative, subject to review. We still need to check the output, especially where a group is small enough to raise identification or disclosure concerns.

B. Correct. A disclosure review remains part of the proposal. Do not treat a change in table shape as an automatic conclusion that all privacy and handling conditions have disappeared.

C. Exactly. Access expiry and data retention are separate questions. The actual retention and deletion arrangements need their own approved basis rather than being inferred from this access duration.

D. Include the data lineage for the aggregate counts as well. That preserves both analytical trust and the handling boundary while giving you the information the actual question needs.

Answers and explanations, page 155

A Read the situation

LESSON 08 / INSIGHT STORYTELLING AND RECOMMENDATIONS

Lead a short briefing with a bounded action, quantified evidence, explicit limits, and a review point.

> **Five minutes to authorize the right investigation**
>
> Analyst Oren has five minutes with operations director Claudia. Two complete weeks use the same ticket definition: issue Q appears in 20 of 1,000 tickets in week one and 80 of 1,000 in week two. Seventy of the later 80 issue-Q tickets are mobile-related; the total mobile-ticket denominator is not supplied. A release occurred between weeks, but cause is unestablished. Oren requests a two-day investigation led by the service analyst, with a readout afterward. Approval is for investigation only, not a production change.

People: Oren, Data analyst; Claudia, Operations director.

Read the record

Choose one answer per question using the facts above.

1. How did the issue-Q ticket rate change?

A. From 8% to 2%, a decline
B. From 2% to 8%, a six-percentage-point rise
C. From 20% to 80%, a sixty-point rise
D. From 2% to 8%, a six-percent relative rise

2. What does the mobile count establish?

A. The mobile failure rate is proven higher than desktop's.
B. Seventy of the eighty later issue-Q tickets are mobile-related.
C. 87.5% of every mobile user experienced the issue.
D. Every mobile ticket concerns Q.

3. What authorization is requested?

A. A two-day investigation with a subsequent readout
B. An immediate unreviewed production change
C. A declaration that the release caused Q
D. Permanent access to all customer records

Answers and explanations, page 156

B Vocabulary in use

CORE TERMS

insight
A decision-relevant interpretation supported by evidence.

state the insight

recommendation
A proposed action with a reasoned basis.

frame the recommendation

decision ask
The precise choice requested from the audience.

lead with the decision ask

headline finding
The main evidence statement in a short briefing.

sharpen the headline finding

supporting evidence
Facts that substantiate a conclusion or recommendation.

present supporting evidence

issue category
A defined class used to group reported problems.

validate the issue category

ticket mix
The composition of support records across categories or groups.

analyze the ticket mix

segment concentration
A large share of observations within a particular subgroup.

report segment concentration

base rate
The underlying frequency relative to an eligible population.

check the base rate

denominator gap
Missing information needed to compute the relevant rate.

disclose the denominator gap

descriptive finding
A statement about observed data without a causal conclusion.

distinguish descriptive findings

causal hypothesis
A proposed explanation of what produced an outcome.

test the causal hypothesis

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

insight INSIGHT

state the insight

Keep the meaning precise: A decision-relevant interpretation supported by evidence.

issue category ISSUE CATEGORY

validate the issue category

Keep the meaning precise: A defined class used to group reported problems.

B Precision and professional shorthand

EXTEND YOUR RANGE

release timing
The timing of a product or system change.
compare release timing

diagnostic follow-up
Targeted analysis to investigate an observed issue.
scope diagnostic follow-up

investigation owner
The role accountable for the defined inquiry.
assign the investigation owner

time box
A specified limit on the duration of work.
set a time box

readout
A presentation of findings and their implications.
schedule the readout

action boundary
The limit of what a decision authorizes.
state the action boundary

production change
A modification to a live operating system or service.
approve a production change

decision log
A record of choices, reasons, and responsible people.
update the decision log

priority criterion
A stated basis for ordering work.
apply the priority criterion

evidence threshold
The support required for a specified conclusion or next action.
define the evidence threshold

leading question
A question that steers toward a preferred answer.
avoid leading questions

alternative explanation
Another plausible account of an observed pattern.
examine alternative explanations

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

causal hypothesis CAUSAL HYPOTHESIS

test the causal hypothesis

Keep the meaning precise: A proposed explanation of what produced an outcome.

readout READOUT

schedule the readout

Keep the meaning precise: A presentation of findings and their implications.

C Phrases that move the work forward

Open with the action
"I recommend a two-day investigation of issue Q."

State the evidence
"The ticket rate rose from 2% to 8%."

Name the unit
"Those rates describe tickets, not unique users."

Give the change
"That is a six-percentage-point increase."

Show the counts
"The counts are 20 of 1,000 and 80 of 1,000."

Identify the concentration
"Seventy of the later eighty issue tickets are mobile-related."

Limit the segment claim
"We do not have the total mobile-ticket denominator."

Keep cause open
"The release timing is a lead, not proof of cause."

Propose the focused check
"Examine affected mobile paths and classification consistency."

Name the owner
"The service analyst would lead the investigation."

Set the duration
"The proposed time box is two days."

Define the output
"Return with evidence, remaining uncertainty, and a next-step recommendation."

Preserve the boundary
"This request does not authorize a production change."

Avoid a discovery promise
"The readout date is not a guaranteed fix date."

Invite the actual decision
"Please approve or decline that investigation scope."

Close the briefing
"The evidence supports investigating now without pretending the cause is settled."

Precision in real exchanges

Purpose: Give the change
Useful wording: "That is a six-percentage-point increase."

Purpose: Show the counts
Useful wording: "The counts are 20 of 1,000 and 80 of 1,000."

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which opening best uses the five-minute meeting?

A. Approve a two-day investigation of Q; its ticket rate rose from 2% to 8%.

B. Here are ten unrelated tables with no action request.

C. The release definitely broke every mobile session.

D. We should change production immediately because the chart rose.

2. What is the relative increase from 2% to 8%?

A. 400%

B. 87.5%

C. 300%

D. 6%

3. A colleague needs the term for this meaning: The main evidence statement in a short briefing. Which term fits?

A. headline finding

B. base rate

C. investigation owner

D. priority criterion

D Choose the response that fits / continued

4. Rebuild the exchange from "Five minutes to authorize the right investigation". Claudia: "That is six percentage points, and four times the earlier rate. We should not call it six-percent growth or accidentally present ticket counts as unique customers." Oren: [missing response] Claudia: "Seventy of the eighty are mobile-related. Can I say mobile fails more often than desktop, or are we missing the denominators for that comparison?" Which response belongs in the gap?

A. Correct. The headline finding is a rise in the issue's ticket rate, not proof of how many distinct people were affected or why the issue increased.

B. The action boundary is investigation only. Any production change would need its separate evidence, review, and authority. We are not asking this meeting to bypass that process.

C. Exactly. Disclose the denominator gap and keep the descriptive finding. Seventy divided by eighty is eighty-seven point five percent of these issue tickets, not of mobile users.

D. Yes. The decision ask is approval of that scope. I will record your decision and preserve the ticket units, missing denominator, and causal limitation in the briefing.

Answers and explanations, page 156

> **One useful distinction**
>
> Issue Q rises from 2% to 8% of tickets: +6 percentage points, four times the earlier rate, or a 300% relative increase. The later mobile share is 70/80 = 87.5% of issue-Q tickets, not of mobile users.

E Conversation

Five minutes to authorize the right investigation

Five minutes to authorize the right investigation

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

decision ask / investigation owner / denominator gap / action boundary / recommendation / headline finding / supporting evidence / time box / causal hypothesis / segment concentration

01 **Claudia:** We have five minutes and ten findings. Which decision needs my attention now, and what evidence supports it?

02 **Oren:** My [1] _____ is a two-day investigation of issue Q. Its share of tickets rose from two percent to eight percent across two complete weeks using the same classification.

03 **Claudia:** Give me the actual counts as well. I want to know whether the percentage changed because the ticket base became much smaller.

04 **Oren:** The [2] _____ is twenty of one thousand tickets, then eighty of one thousand. The bases are equal, so the rate increase is not caused by a smaller denominator here.

05 **Claudia:** That is six percentage points, and four times the earlier rate. We should not call it six-percent growth or accidentally present ticket counts as unique customers.

06 **Oren:** Correct. The [3] _____ is a rise in the issue's ticket rate, not proof of how many distinct people were affected or why the issue increased.

E Conversation / continued

07 **Claudia:** Seventy of the eighty are mobile-related. Can I say mobile fails more often than desktop, or are we missing the denominators for that comparison?

08 **Oren:** No. That is [4] _____ within the issue tickets. We lack the total mobile-ticket denominator, so we cannot calculate the relevant mobile rate or compare it fairly with desktop.

09 **Claudia:** But the concentration still gives the investigator a useful place to start. We can inspect the mobile paths without overstating what seventy out of eighty means.

10 **Oren:** Exactly. Disclose the [5] _____ and keep the descriptive finding. Seventy divided by eighty is eighty-seven point five percent of these issue tickets, not of mobile users.

Conversation continues on the next page.

Five minutes to authorize the right investigation

CONVERSATION / CONTINUED

11 **Claudia:** A release occurred between the two weeks. That timing sounds relevant, although the current report does not show whether the release actually caused the increase.

12 **Oren:** Treat it as a [6] _____. The investigation should examine that lead and alternatives, including whether classification or affected paths changed. The sequence alone does not establish cause.

13 **Claudia:** Who would lead the work? I want a named role and a defined output, not an open request for somebody to analyze the problem someday.

14 **Oren:** The service analyst is the proposed [7] _____. The output is a readout of evidence, remaining uncertainty, and a recommended next step, rather than an unsupported promise of a fix.

15 **Claudia:** Two days for an evidence readout, not a guaranteed root cause. Please make that explicit in the invitation before people start expecting a fix.

16 **Oren:** It is a [8] _____, not a certainty deadline. We should report what the inquiry establishes and what remains unresolved at the readout, even if the original hypothesis is not supported.

17 **Claudia:** Does approving this request also allow the analyst to change production if a likely cause appears? I want that distinction written clearly before I decide.

Five minutes to authorize the right investigation / continued

18 **Oren:** The [9] _____ is investigation only. Any production change would need its separate evidence, review, and authority. We are not asking this meeting to bypass that process.

19 **Claudia:** Then the decision is whether to allocate two days to a documented service issue, with the mobile concentration as a lead and the cause still open.

20 **Oren:** Yes. The [10] _____ is approval of that scope. I will record your decision and preserve the ticket units, missing denominator, and causal limitation in the briefing.

CHECK THE GAPS

1. recommendation / **2.** supporting evidence / **3.** headline finding / **4.** segment concentration / **5.** denominator gap / **6.** causal hypothesis / **7.** investigation owner / **8.** time box / **9.** action boundary / **10.** decision ask

Why these answers fit, page 157

F Check accuracy and scope

Five minutes to authorize the right investigation

Return to the case

Analyst Oren has five minutes with operations director Claudia. Two complete weeks use the same ticket definition: issue Q appears in 20 of 1,000 tickets in week one and 80 of 1,000 in week two. Seventy of the later 80 issue-Q tickets are mobile-related; the total mobile-ticket denominator is not supplied. A release occurred between weeks, but cause is unestablished. Oren requests a two-day investigation led by the service analyst, with a readout afterward. Approval is for investigation only, not a production change.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why is 70/80 not a mobile failure rate?

A. Seventy divided by eighty cannot be calculated.
B. Every issue ticket is a unique mobile user by definition.
C. The earlier week had no mobile users.
D. Its denominator is issue-Q tickets, not all relevant mobile activity.

2. What does approval of this request authorize?

A. Any immediate production change the analyst prefers
B. A public claim that the release caused the issue
C. An indefinite project without review
D. The defined two-day investigation and readout

Answers and explanations, page 159

Listen for the distinction

Switch roles for turns 11-20. Stress investigation only and the two-day readout.

Make uncertainty actionable

A concise recommendation does not need an invented root cause. State the observed change, why it merits attention, and the smallest defined evidence step that supports the next decision. Acknowledge what a concentration of tickets does not tell you about the wider user population.

F Say it / transfer to a new case

A concentration is a lead, not a rate

Issue R appears in 10 of 500 tickets, then 30 of 500. Twenty-four of the later 30 issue tickets concern mobile. The total mobile base is unknown. The request is a one-day investigation, not a live change.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

80 / four / six / investigation

01 **Analyst:** The later issue rate is [1] _____ percent.

02 **Director:** The increase is [2] _____ percentage points.

03 **Analyst:** Mobile represents [3] _____ percent of later issue tickets.

04 **Director:** Approval would authorize an [4] _____.

CHECK THE GAPS

1. six / **2.** four / **3.** 80 / **4.** investigation

Why these answers fit, page 159

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Five minutes to authorize the right investigation". Claudia: "Who would lead the work? I want a named role and a defined output, not an open request for somebody to analyze the problem someday." Oren: [missing response] Claudia: "Two days for an evidence readout, not a guaranteed root cause. Please make that explicit in the invitation before people start expecting a fix." Which response belongs in the gap?

A. The supporting evidence is twenty of one thousand tickets, then eighty of one thousand. The bases are equal, so the rate increase is not caused by a smaller denominator here.

B. The service analyst is the proposed investigation owner. The output is a readout of evidence, remaining uncertainty, and a recommended next step, rather than an unsupported promise of a fix.

C. It is a time box, not a certainty deadline. We should report what the inquiry establishes and what remains unresolved at the readout, even if the original hypothesis is not supported.

D. No. That is segment concentration within the issue tickets. We lack the total mobile-ticket denominator, so we cannot calculate the relevant mobile rate or compare it fairly with desktop.

Answers and explanations, page 160

Review 4 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Data Governance and Privacy" and "Insight Storytelling and Recommendations". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: The role accountable for specified data decisions. Which term fits?

A. indirect identifier

B. data owner

C. data catalog

D. disclosure review

2. A colleague needs the term for this meaning: A large share of observations within a particular subgroup. Which term fits?

A. segment concentration

B. release timing

C. insight

D. production change

3. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "Then should the slide say no effect? Zero is inside the interval, but so are positive effects. I do not want to replace one overclaim with another." Theo: [missing response] Hana: "Then the decision is clearer than the underlying effect. We know our agreed rollout rule failed, while the magnitude and even direction remain uncertain." Which response belongs in the gap?

A. No. The no-change value is included, but useful positive effects are included too. The readout does not establish exact zero or show that the two experiences are equivalent.

B. We reached the planned endpoint. Continuing until the result looks favorable changes the statistical behavior. There is no authorized extension or approved repeated-decision method in the supplied plan.

C. Exactly. Apply the whole decision rule. We can acknowledge the guardrail result while saying the test did not authorize rollout under the standard agreed before seeing these results.

D. Yes. An appropriate sequential analysis can support planned repeated decisions, but that is not what was specified here. Any new evaluation needs its own reviewed assumptions and rules.

Review 4 / Meaning and context / continued

4. Rebuild the exchange from "Five minutes to authorize the right investigation". Claudia: "Does approving this request also allow the analyst to change production if a likely cause appears? I want that distinction written clearly before I decide." Oren: [missing response] Claudia: "Then the decision is whether to allocate two days to a documented service issue, with the mobile concentration as a lead and the cause still open." Which response belongs in the gap?

A. Yes. The decision ask is approval of that scope. I will record your decision and preserve the ticket units, missing denominator, and causal limitation in the briefing.

B. My recommendation is a two-day investigation of issue Q. Its share of tickets rose from two percent to eight percent across two complete weeks using the same classification.

C. Exactly. Disclose the denominator gap and keep the descriptive finding. Seventy divided by eighty is eighty-seven point five percent of these issue tickets, not of mobile users.

D. The action boundary is investigation only. Any production change would need its separate evidence, review, and authority. We are not asking this meeting to bypass that process.

Answers and explanations, page 170

Language for the next case

Both segments improve, but the overall rate falls

denominator

The denominator is the eligible visit count for the corresponding segment and period; those counts change substantially even though each period totals one thousand visits.

Nadia: They use the same records. Start with each denominator. A changes from one hundred to nine hundred visits; B changes from nine hundred to one hundred.

percentage points

Percentage points express the difference between the percentage values; both segment rates rise by two points, not by the same relative percentage.

Nadia: Yes, up two percentage points. B goes from thirty percent to thirty-two percent. Both segment-level comparisons are favorable.

mix shift

The mix shift moves a much larger share of visits into lower-converting segment A; changing weights can reverse the direction seen in the separate segment rates.

Nadia: Exactly. This mix shift explains why the aggregate comparison moves differently from the segment comparisons. An unweighted average would answer another question.

reference weights

The reference weights are the earlier ten-percent share for A and ninety-percent share for B; keeping them fixed creates a separate composition-standardized comparison.

Nadia: Yes, if we label the calculation. Use the earlier reference weights: ten percent for A and ninety percent for B.

standardization

Standardization applies a common weighting basis to improve comparability; it does not prove that an intervention changing traffic would leave every segment's rate unchanged.

Nadia: That overstates the standardization. It is a descriptive comparison, not proof that a future allocation change would preserve those rates.

causal effect

A causal effect concerns what an intervention changes; these aggregate and standardized comparisons alone do not isolate the effect of changing traffic allocation.

Nadia: Yes. We have not estimated a causal effect of reallocating traffic. The arithmetic explains the apparent contradiction without settling the intervention decision.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

Both segments improve, but the overall rate falls

Two periods each contain 1,000 eligible visits. Segment A changes from 10 conversions in 100 visits to 108 in 900. Segment B changes from 270 in 900 to 32 in 100. Definitions are unchanged. The team must explain segment improvement alongside an overall decline.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

denominator / reference weights / mix shift / standardization / causal effect / percentage points

01 **Paul:** Your segment chart shows improvement in both groups, but the headline says conversion fell. Have the charts used different data?

02 **Nadia:** They use the same records. Start with each [1] _____. A changes from one hundred to nine hundred visits; B changes from nine hundred to one hundred.

03 **Paul:** A goes from ten out of one hundred to one hundred eight out of nine hundred. That is ten percent to twelve percent.

04 **Nadia:** Yes, up two [2] _____. B goes from thirty percent to thirty-two percent. Both segment-level comparisons are favorable.

05 **Paul:** The first total is ten plus two hundred seventy conversions: two hundred eighty out of one thousand, or twenty-eight percent.

06 **Nadia:** And the later total is one hundred eight plus thirty-two: one hundred forty out of one thousand, or fourteen percent. Those calculations reconcile.

Conversation / continued

07 **Paul:** So the later period has many more visits in the lower-converting segment. Its better rate does not offset that change in composition.

08 **Nadia:** Exactly. This [3] _____ explains why the aggregate comparison moves differently from the segment comparisons. An unweighted average would answer another question.

09 **Paul:** Could we just show the average of the two segment rates? That would be twenty percent before and twenty-two percent after.

10 **Nadia:** Only with an explicit equal-weight interpretation. It would not be the actual share of these visits that converted, because the segment sizes are not equal.

Conversation continues on the next page.

Both segments improve, but the overall rate falls

CONVERSATION / CONTINUED

11 **Paul:** For a like-for-like composition comparison, could we apply the earlier visit proportions to the later segment rates?

12 **Nadia:** Yes, if we label the calculation. Use the earlier [4] _____: ten percent for A and ninety percent for B.

13 **Paul:** That gives point one times twelve percent plus point nine times thirty-two percent. The result is thirty percent, not the observed fourteen percent.

14 **Nadia:** Correct. It shows what that weighted calculation produces using the earlier mix. Keep the observed result alongside it; do not replace actual performance with the adjusted figure.

15 **Paul:** The commercial team might call thirty percent the performance we would certainly get if we changed our traffic allocation back.

16 **Nadia:** That overstates the [5] _____. It is a descriptive comparison, not proof that a future allocation change would preserve those rates.

17 **Paul:** Then we still need evidence about why the mix changed and whether different visitors within each segment would behave similarly.

18 **Nadia:** Yes. We have not estimated a [6] _____ of reallocating traffic. The arithmetic explains the apparent contradiction without settling the intervention decision.

19 **Paul:** I will show actual conversion, segment rates, and segment sizes together. The standardized comparison can be separate and clearly labeled.

20 **Nadia:** That lets the audience see both the real aggregate decline and the improvement within each segment, without choosing whichever chart tells the preferred story.

CHECK THE GAPS

1. denominator / **2.** percentage points / **3.** mix shift / **4.** reference weights / **5.** standardization / **6.** causal effect

Why these answers fit, page 173

An equal-weight average hides the actual visit mix

Earlier: A converts 20 of 100 visits; B converts 180 of 300. Later: A converts 90 of 300; B converts 70 of 100. Definitions match. Each period has 400 visits.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

50 / 40 / ten / composition

01 **Analyst:** The earlier overall rate is [1] _____ percent.

02 **Director:** The later overall rate is [2] _____ percent.

03 **Analyst:** Each segment improved by [3] _____ percentage points.

04 **Director:** Show the segment sizes so the changing [4] _____ is visible.

CHECK THE GAPS

1. 50 / **2.** 40 / **3.** ten / **4.** composition

Why these answers fit, page 173

Keep the distinction audible

Explain weighted rates and changing composition without inventing a causal explanation.

Language for the next case

The customer moved regions. Did the old sale move too?

current record

The current record contains the customer's latest region, South; using it for every order reclassifies earlier sales instead of preserving their original regional attribution.

Iris: Yes. The report joins both orders to the current record. It now says South, so the February order moves with the customer in that view.

historical attribution

Historical attribution assigns each fact using the relevant state at its event time; it differs from regrouping all facts under today's customer attributes.

Iris: Then we need a historical attribution rule. Neither view should be labeled simply sales by region without telling readers which basis it uses.

Type 2 dimension

A Type 2 dimension preserves successive versions of descriptive attributes so facts can be related to the version valid at the relevant time.

Iris: Two versions can represent one business customer. A Type 2 dimension preserves the region history instead of overwriting every previous state.

surrogate key

A surrogate key identifies the particular warehouse dimension row; the stable customer business identifier can appear on several historical versions.

Iris: Each version gets a surrogate key. Facts can then reference the correct version while we still group those versions under the same customer when needed.

time-based lookup

A time-based lookup matches the fact's relevant date to the dimension version's effective interval, rather than selecting the current version for every fact.

Iris: Use a time-based lookup against the effective interval for C17. Then test the boundary date, missing intervals, and overlapping versions.

restatement

Restatement revises previously reported figures under a documented correction policy; the changed attribution and its reason must remain traceable.

Iris: Follow the agreed restatement policy. Record the corrected source evidence, affected periods, and what changed in the published figures.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The customer moved regions. Did the old sale move too?

Customer C17 belonged to North in February and moved to South effective March 1. Its February order is $500; its April order is $700. A current-region join assigns both to South. The requested report must use the region in effect on each order date.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

historical attribution / time-based lookup / surrogate key / Type 2 dimension / restatement / current record

01 **Mateo:** South has gained five hundred dollars in February, even though nobody entered a new order. Could the customer-region update explain it?

02 **Iris:** Yes. The report joins both orders to the [1] _____. It now says South, so the February order moves with the customer in that view.

03 **Mateo:** Our requested view is region at the time of sale. February should remain North and April should be South.

04 **Iris:** Then we need a [2] _____ rule. Neither view should be labeled simply sales by region without telling readers which basis it uses.

05 **Mateo:** We have the approved March first effective date. Does the warehouse need two customer records, or would that duplicate the customer?

06 **Iris:** Two versions can represent one business customer. A [3] _____ preserves the region history instead of overwriting every previous state.

07 **Mateo:** C17 remains the business identifier in both versions. How do we distinguish the rows if that identifier is no longer unique in the history table?

08 **Iris:** Each version gets a [4] _____. Facts can then reference the correct version while we still group those versions under the same customer when needed.

09 **Mateo:** For the boundary, the North version ends when the South version starts. We must avoid matching the March first order to both.

10 **Iris:** Exactly. Define the effective intervals consistently. In this model the start is inclusive and the end exclusive, so March first belongs to South only.

Conversation continues on the next page.

The customer moved regions. Did the old sale move too?

CONVERSATION / CONTINUED

11 **Mateo:** What does the loading process use to assign the version to each order? A current flag would give the same wrong answer again.

12 **Iris:** Use a [5] _____ against the effective interval for C17. Then test the boundary date, missing intervals, and overlapping versions.

13 **Mateo:** The expected result for these two orders is North five hundred, South seven hundred, total twelve hundred. The total alone would not expose the original error.

14 **Iris:** Right. The current-region report also totals twelve hundred. Check the regional allocation as well as the grand total, and retain both labeled views if both are required.

15 **Mateo:** Suppose the source team later corrects the move date. We should not silently replace an already issued regional report without a record.

16 **Iris:** Follow the agreed [6] _____ policy. Record the corrected source evidence, affected periods, and what changed in the published figures.

17 **Mateo:** Could we infer the old region from today's address if the earlier version is missing? That would make the chart complete.

18 **Iris:** Not without supporting evidence. Flag the unresolved history and route it to the source owner. A complete-looking chart is not a reason to invent an earlier state.

19 **Mateo:** I will confirm the requested historical basis with Sales, then check the two orders and the March first boundary in the revised output.

20 **Iris:** I will preserve C17's business identity, assign the appropriate version keys, and document the interval rule so future refreshes do not silently move old sales.

CHECK THE GAPS

1. current record / **2.** historical attribution / **3.** Type 2 dimension / **4.** surrogate key / **5.** time-based lookup / **6.** restatement

Why these answers fit, page 175

The grand total passes while the regional split is wrong

Customer D8 moves from East to West effective May 1. It has a $300 April order and a $400 June order. The report requires order-date region, not current region.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

West / 700 / East / keys

01 **Analyst:** The April order belongs to [1] _____.

02 **Engineer:** The June order belongs to [2] _____.

03 **Analyst:** The unchanged grand total is [3] _____ dollars.

04 **Engineer:** Verify the effective-date intervals and the selected version [4] _____.

CHECK THE GAPS

1. East / **2.** West / **3.** 700 / **4.** keys

Why these answers fit, page 175

Keep the distinction audible

Agree historical attribution and distinguish business identity from a versioned dimension record.

Language for the next case

The forecast used information from tomorrow

forecast origin

The forecast origin is the time when the prediction is issued; information generated afterward cannot be treated as an available input to that prediction.

Farah: One input was unavailable at the forecast origin. It used tomorrow's resolved-ticket count to predict tomorrow's incoming volume.

data leakage

Data leakage introduces information unavailable for the intended prediction into model development or evaluation, making the reported performance unrepresentative of the real forecasting task.

Farah: Exactly. That is data leakage. A field being present in today's historical file does not mean it existed at the earlier decision time.

backtest

A backtest evaluates predictions against historical outcomes while recreating the information and timing of the intended forecasting task.

Farah: We used a chronological backtest. Training precedes the evaluated predictions, and each forecast uses only the information available at its issue time.

holdout period

The holdout period is reserved for evaluation and was not used to fit or tune this model; repeatedly tuning against it would undermine that separation.

Farah: No. It is the fixed holdout period. The model and comparison rule were settled before those results were examined.

mean absolute error

Mean absolute error averages the magnitudes of prediction errors without cancellation; eighty tickets of total absolute error over ten days gives eight tickets per daily forecast.

Farah: The mean absolute error is eight tickets per daily forecast. It is in ticket units, not eight percent, and opposite signed errors do not cancel.

baseline

The baseline is the agreed reference forecasting method evaluated on the same days and outcome definition; it gives the new model's error a meaningful comparison.

Farah: Yes. That baseline is essential. A model can look sophisticated while failing to improve on a simple alternative.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The forecast used information from tomorrow

An old next-day ticket-volume model used next-day resolved-ticket counts, unavailable when forecasting. A separate rebuilt model uses archived, time-appropriate inputs. On a fixed ten-day holdout unused for training or tuning, its total absolute error is 80 tickets; the agreed baseline totals 100.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

mean absolute error / data leakage / holdout period / forecast origin / backtest / baseline

01 **Lewis:** The old forecast was remarkably accurate. Why have you taken its performance figure out of the planning presentation?

02 **Farah:** One input was unavailable at the [1] _____. It used tomorrow's resolved-ticket count to predict tomorrow's incoming volume.

03 **Lewis:** That count was in the historical export, so it looked like an ordinary column. But we would not have known it when making the forecast.

04 **Farah:** Exactly. That is [2] _____. A field being present in today's historical file does not mean it existed at the earlier decision time.

05 **Lewis:** You rebuilt the model without it. Have you also checked when the remaining fields became available, rather than just when their events occurred?

06 **Farah:** Yes. This rebuilt evaluation uses archived inputs available at each prediction time, including the actual publication delays. It does not backdate a later data delivery.

07 **Lewis:** How did you test it? A random mixture of past and future dates would not look like our daily planning process.

08 **Farah:** We used a chronological [3] _____. Training precedes the evaluated predictions, and each forecast uses only the information available at its issue time.

09 **Lewis:** The comparison sheet also names a final ten-day block. Was that block used to choose the model settings?

10 **Farah:** No. It is the fixed [4] _____. The model and comparison rule were settled before those results were examined.

Conversation continues on the next page.

The forecast used information from tomorrow

CONVERSATION / CONTINUED

11 **Lewis:** Across those ten days the absolute errors add to eighty tickets. What should I call the average, rather than just saying eighty errors?

12 **Farah:** The [5] _____ is eight tickets per daily forecast. It is in ticket units, not eight percent, and opposite signed errors do not cancel.

13 **Lewis:** Our agreed simple forecasting method has total absolute error of one hundred over the same ten days, so its mean is ten tickets.

14 **Farah:** Yes. That [6] _____ is essential. A model can look sophisticated while failing to improve on a simple alternative.

15 **Lewis:** Eight instead of ten is a twenty-percent reduction in mean absolute error for this period. It does not mean eighty-percent forecast accuracy.

16 **Farah:** Correct. Also inspect individual days and the operational cost of underestimation versus overestimation. The mean alone does not describe every planning consequence.

17 **Lewis:** Can we promise the same improvement next month? Staffing needs a practical expectation, not just the historical result.

18 **Farah:** No guarantee. This is a bounded historical evaluation. Changes in demand or input availability can alter performance, so the approved deployment plan needs monitoring.

19 **Lewis:** I will present the eight-versus-ten-ticket comparison, name the ten-day period, and explain why the earlier figure was withdrawn.

20 **Farah:** Good. Keep the corrected evaluation traceable and the future claim modest. The result supports considering the rebuilt model, not pretending we can use tomorrow's information today.

CHECK THE GAPS

1. forecast origin / **2.** data leakage / **3.** backtest / **4.** holdout period / **5.** mean absolute error / **6.** baseline

Why these answers fit, page 177

A ticket error is mislabeled as a percentage

Over eight held-out days, a model has 48 tickets of total absolute error. The reference method has 64 over the same days. No percentage-error metric was calculated.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

eight / error / six / twenty-five

01 **Analyst:** The model mean absolute error is [1] _____ tickets per forecast.

02 **Planner:** The reference method averages [2] _____ tickets.

03 **Analyst:** That is a [3] _____ percent reduction in this error metric.

04 **Planner:** Do not relabel a six-ticket error as six-percent [4] _____.

CHECK THE GAPS

1. six / **2.** eight / **3.** twenty-five / **4.** error

Why these answers fit, page 177

Keep the distinction audible

Challenge an evaluation shortcut and explain forecast error in the correct units.

01 / Answers and explanations

A / Read the situation

A1. Which count answers the supplied business question?

A. 900 distinct users with a completed transaction

The question concerns eligible users completing transactions, which the second measure directly counts.

Return to the exercise, page 4

A2. What is the correct eligible-user transaction rate?

C. 45%

Nine hundred divided by two thousand eligible users equals forty-five percent.

Return to the exercise, page 4

A3. Why can both query results be correct?

D. They apply different event definitions to the same eligible population.

Distinct event definitions can produce different accurate totals without a calculation error.

Return to the exercise, page 4

D / Response choices

D1. Which label best fits the 900 count?

A. September transaction-active users

The count concerns distinct users with a completed transaction in the reporting window.

Return to the exercise, page 8

D2. What does 900/1,200 measure?

B. The share of logged-in users who completed a transaction

The denominator is the login group, and the numerator is its transacting subset.

Return to the exercise, page 8

01 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: An action meeting the condition for inclusion. Which term fits?

C. qualifying event

A Not here. test account: An account used for testing rather than genuine activity. The usual combination is "filter test accounts".

B Not here. numerator: The counted outcome placed over the base of a rate. The usual combination is "verify the numerator".

C Correct. qualifying event: An action meeting the condition for inclusion. The usual combination is "specify the qualifying event".

D Not here. subset: A group entirely contained within another group. The usual combination is "identify the subset relationship".

Return to the exercise, page 8

D4. Rebuild the exchange from "Two active-user totals, two different events". Rafael: "The window is the September calendar month in UTC for both, and they use the same two thousand eligible users. That part is aligned." Mei: [missing response] Rafael: "For the actual question, I should choose nine hundred and label it as people completing a transaction. Multiple transactions by one user still count once." Which response belongs in the gap?

B. Good. A shared reporting window removes one possible source of disagreement. The distinction here is activity type, not different month boundaries or different treatment of test accounts.

A Not here. This line belongs to turn 10, after Rafael says: "I have seventy-five percent on the slide: nine hundred divided by twelve hundred. Is that answering the eligible-user question, or have I changed the base?" It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Mei's response in turn 6. It connects Rafael's preceding point with Rafael's follow-up. The reporting window fixes the period and prevents a time-boundary difference from being confused with the event-definition difference.

C Not here. This line belongs to turn 18, after Rafael says: "Who signs off if the transaction event changes next month? I do not want the same label to hide a new definition." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Rafael says: "My summary will be nine hundred transaction-active users, forty-five percent of two thousand eligible users, September UTC. The login measure remains separately labeled." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 8

01 / Answers and explanations / continued

E / Conversation

1. business question

Turn 2. The business question determines which correctly calculated measure is relevant to the decision.

2. qualifying event

Turn 4. The qualifying event differs between successful login and completed transaction, explaining the distinct totals.

3. reporting window

Turn 6. The reporting window fixes the period and prevents a time-boundary difference from being confused with the event-definition difference.

4. distinct user

Turn 8. A distinct user is counted once even when the user performs multiple qualifying actions within the window.

5. denominator

Turn 10. The denominator defines the rate's base; eligible users and logged-in users answer different percentage questions.

6. numerator

Turn 12. The numerator remains nine hundred transacting users while the stated denominator changes the meaning of the percentage.

7. subset

Turn 14. The subset relationship means every transacting user is already counted within the login group, so addition would duplicate those users.

8. semantic layer

Turn 16. The semantic layer preserves shared business definitions so later reports use the same labels and rules consistently.

9. metric owner

Turn 18. The metric owner is accountable for the definition and its controlled evolution rather than leaving each report to invent its own meaning.

01 / Answers and explanations / continued

10. reconciliation

Turn 20. Reconciliation explains the different totals and their uses without forcing distinct measures to produce an artificial match.

Return to the conversation, page 10

F / Accuracy and scope

F1. Why should the two totals not be added?

B. The 900 transacting users are already included among the 1,200 login users.

The supplied subset relationship means adding totals double-counts the transacting group.

Return to the exercise, page 14

F2. Which next step addresses the real disagreement?

D. Record both event definitions and select the transaction measure for this question.

The discrepancy is semantic and decision-related, not a demonstrated query failure.

Return to the exercise, page 14

F / Guided speaking

1. 400

Analyst: The relevant distinct-user count is 400.

Turn 1. The supplied transaction-active group contains four hundred distinct users.

2. 1000

Manager: The eligible-user denominator is 1000.

Turn 2. The requested rate uses all one thousand eligible users as its base.

3. 40

Analyst: The requested rate is 40 percent.

Turn 3. Four hundred divided by one thousand equals forty percent.

4. 50

Manager: Using login users instead would give 50 percent.

Turn 4. Four hundred divided by eight hundred logged-in users equals fifty percent.

Return to the conversation, page 15

01 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Two active-user totals, two different events". Rafael: "Someone suggested adding the two reports to show total engagement. But the nine hundred transacting users are already included in the twelve hundred who logged in." Mei: [missing response] Rafael: "I will keep the login figure because it is useful, but give it a different label. The executive headline will use the transaction definition." Which response belongs in the gap?

C. That subset relationship makes addition inappropriate for distinct people. It would count the transaction group twice rather than identify twenty-one hundred different active users.

A Not here. This line belongs to turn 2, after Rafael says: "The executive slide says twelve hundred active users, but Finance has nine hundred. Both teams insist their query is right. Which number should I remove?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Rafael says: "Both rates tell us something, but forty-five percent is the answer to the eligible-user question. Seventy-five percent describes the logged-in group instead." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Mei's response in turn 14. It connects Rafael's preceding point with Rafael's follow-up. The subset relationship means every transacting user is already counted within the login group, so addition would duplicate those users.

D Not here. This line belongs to turn 20, after Rafael says: "My summary will be nine hundred transaction-active users, forty-five percent of two thousand eligible users, September UTC. The login measure remains separately labeled." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 15

02 / Answers and explanations

A / Read the situation

A1. Which source is complete only through Monday?

A. Refunds

The briefing explicitly limits refund coverage to Monday while orders extend through Tuesday.

Return to the exercise, page 16

A2. What is known about Tuesday refunds?

D. They are unknown in the available source.

Missing source coverage does not establish the amount of Tuesday refunds.

Return to the exercise, page 16

A3. Which comparison can Asha publish as verified?

C. Monday-only, using aligned source coverage

Both sources are verified complete through Monday, making that shared period usable.

Return to the exercise, page 16

D / Response choices

D1. Which dashboard note is accurate?

A. Refreshed Wednesday 09:00 UTC; refunds complete through Monday; two-day net value incomplete.

The note separates successful processing from source coverage and identifies the affected measure.

Return to the exercise, page 20

D2. Why is Monday-only reporting usable here?

D. Both sources are verified complete through that shared period.

The relevant fact is aligned verified coverage, not an inherent property of Monday.

Return to the exercise, page 20

02 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Presence of the required records or values. Which term fits?

B. completeness

A Not here. null value: A marker for missing or unknown information in a data field. The usual combination is "distinguish null from zero".

B Correct. completeness: Presence of the required records or values. The usual combination is "verify source completeness".

C Not here. refresh timestamp: The time a report or model was last updated. The usual combination is "display the refresh timestamp".

D Not here. ingestion delay: A lag in receiving data from a source. The usual combination is "investigate ingestion delay".

Return to the exercise, page 20

D4. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "Then the two-day net figure has Tuesday orders but not the corresponding Tuesday refunds. It could look better simply because that input is incomplete." Asha: [missing response] Tomas: "The chart fills the empty Tuesday cell with zero. Should we change that to unknown while the refund delivery is missing?" Which response belongs in the gap?

B. Yes. The coverage window is misaligned. We cannot interpret the result as a complete two-day net amount when the subtraction uses refunds from a shorter period.

A Not here. This line belongs to turn 4, after Tomas says: "I had assumed that a successful refresh meant every table contained the latest business activity. Which boundary should I read instead for the refund data?" It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Asha's response in turn 6. It connects Tomas's preceding point with Tomas's follow-up. The coverage window differs across orders and refunds, making the combined two-day net measure misaligned.

C Not here. This line belongs to turn 2, after Tomas says: "The dashboard says refreshed this morning. Can I use the Monday-Tuesday net-order-value number in the operations meeting as our complete two-day result?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Tomas says: "I still need a reliable number for the meeting. Could we show Monday by itself, since both sources are verified complete through that day?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 20

E / Conversation

1. refresh timestamp

Turn 2. The refresh timestamp shows when the dashboard processing completed, not the complete-through date of each input.

02 / Answers and explanations / continued

2. source cutoff

Turn 4. The source cutoff identifies the period through which the refund source is verified complete, here Monday 23:59 UTC.

3. coverage window

Turn 6. The coverage window differs across orders and refunds, making the combined two-day net measure misaligned.

4. zero value

Turn 8. A zero value asserts a quantity of none; absent Tuesday refund coverage does not support that assertion.

5. completeness

Turn 10. Completeness concerns the required records for the stated period; the briefing verifies both sources for Monday.

6. data-quality notice

Turn 12. The data-quality notice makes the source limitation and affected measure visible where readers interpret the result.

7. recovery estimate

Turn 14. A recovery estimate needs a supported expectation; this case supplies no confirmed time for the missing delivery.

8. event time

Turn 16. Event time records when the business event occurred, which remains distinct from the later ingestion or load time.

9. backfill

Turn 18. The backfill loads previously missing records, which need verification before the incomplete-status notice can be removed.

10. timeliness

Turn 20. Timeliness is availability within the required period; the refund delay should be disclosed while the usable aligned view is preserved.

Return to the conversation, page 22

02 / Answers and explanations / continued

F / Accuracy and scope

F1. Which check is insufficient by itself?

A. Seeing that the dashboard refresh job completed

A completed refresh can process stale inputs without repairing their missing coverage.

Return to the exercise, page 25

F2. What should happen after the delayed records arrive?

A. Validate and reconcile them before removing the limitation.

Receiving records is followed by verification, and event dates should preserve business timing.

Return to the exercise, page 25

F / Guided speaking

1. Friday

Analyst: The refresh ran on Friday.

Turn 1. The briefing explicitly identifies Friday as the processing day.

2. Wednesday

Manager: Returns are complete only through Wednesday.

Turn 2. The source cutoff for returns is Wednesday, not the later refresh day.

3. unknown

Analyst: Thursday returns are unknown.

Turn 3. Missing Thursday coverage does not prove a zero business amount.

4. aligned

Manager: A Wednesday-only comparison is aligned.

Turn 4. Both sources are verified complete for Wednesday, so their coverage aligns.

Return to the conversation, page 26

02 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "Do we know when the missing refund delivery will arrive? I would like to tell the meeting whether the full figure will be available this afternoon." Asha: [missing response] Tomas: "Once the rows arrive, we will have business events from Tuesday loaded on a later day. We should not move those refunds into Wednesday totals." Which response belongs in the gap?

C. There is no confirmed recovery estimate. I will request an update from the source owner, but I should not invent a delivery time simply because the meeting needs one.

A Not here. This line belongs to turn 2, after Tomas says: "The dashboard says refreshed this morning. Can I use the Monday-Tuesday net-order-value number in the operations meeting as our complete two-day result?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 18, after Tomas says: "When the delivery arrives, who checks that it is complete? I do not want the warning removed just because the refresh turns green." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Asha's response in turn 14. It connects Tomas's preceding point with Tomas's follow-up. A recovery estimate needs a supported expectation; this case supplies no confirmed time for the missing delivery.

D Not here. This line belongs to turn 12, after Tomas says: "The note should be near the result, not somewhere people will miss it. I will explain that Tuesday orders alone do not settle the net figure." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 26

03 / Answers and explanations

A / Read the situation

A1. What is the September timely-completion rate?

A. 96%

Four hundred eighty timely requests divided by five hundred eligible requests equals ninety-six percent.

Return to the exercise, page 29

A2. How does September compare with the target?

B. One percentage point above the 95% target

Ninety-six percent minus ninety-five percent is one percentage point.

Return to the exercise, page 29

A3. What was the August rate?

D. 90%

Four hundred fifty timely requests divided by five hundred eligible requests equals ninety percent.

Return to the exercise, page 29

D / Response choices

D1. Which card text is most complete?

C. September: 480/500 within 24 hours, 96%; target 95%; August 90%.

The complete text provides the definition, count, denominator, target, and comparable prior rate.

Return to the exercise, page 33

D2. Which change statement is accurate?

D. Up 6 percentage points from August

Ninety-six percent minus ninety percent is six percentage points.

Return to the exercise, page 33

03 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The observed value of the defined measure. Which term fits?

C. actual result

A Not here. drill-down: Moving from a summary to more detailed data. The usual combination is "provide a drill-down".

B Not here. threshold: A boundary used for a rule or decision. The usual combination is "apply the threshold".

C Correct. actual result: The observed value of the defined measure. The usual combination is "report the actual result".

D Not here. month-on-month: Compared with the immediately preceding month. The usual combination is "describe month-on-month movement".

Return to the exercise, page 33

D4. Rebuild the exchange from "Four hundred eighty needs a denominator and a target". Elena: "Our approved target is at least ninety-five percent. That means September passed, but I want the amount above the target described accurately." Jonas: [missing response] Elena: "August had four hundred fifty timely requests out of five hundred. The same definition and exclusions apply, so that earlier rate was ninety percent." Which response belongs in the gap?

A. It is one percentage point above target. Saying one percent without qualification can suggest a relative calculation, which is a different way of expressing the comparison.

A Correct. This is Jonas's response in turn 6. It connects Elena's preceding point with Elena's follow-up. A percentage point expresses the difference between percentage values, so 96% exceeds 95% by one point.

B Not here. This line belongs to turn 18, after Elena says: "The clock begins on receipt and stops on completion. We should also state that the data covers fully observed requests, not a partial month of immature outcomes." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Elena says: "Fair. We can acknowledge the result without attributing it to a particular staffing change that has not been evaluated. The definition should stay visible too." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Elena says: "The card says four hundred eighty with a green arrow. What does it count, and how should I interpret it?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 33

E / Conversation

1. KPI

Turn 2. The KPI is the defined performance measure, here the share of eligible requests completed within twenty-four hours.

03 / Answers and explanations / continued

2. actual result

Turn 4. The actual result is ninety-six percent, calculated from 480 timely completions among 500 eligible requests.

3. percentage point

Turn 6. A percentage point expresses the difference between percentage values, so 96% exceeds 95% by one point.

4. comparison period

Turn 8. The comparison period is the earlier month, whose ninety-percent rate provides the reference for the reported improvement.

5. relative change

Turn 10. Relative change divides the six-point rise by the ninety-percent August rate, giving approximately 6.67%.

6. drill-down

Turn 12. A drill-down exposes the individual or grouped exceptions beneath the favorable summary rate.

7. month-on-month

Turn 14. Month-on-month describes the immediate September-versus-August comparison without claiming a longer established pattern.

8. comparability

Turn 16. Comparability rests on the shared definition, eligibility, and observation basis that make the two monthly rates meaningful to compare.

9. observation window

Turn 18. The observation window must allow the outcome to be observed; the briefing confirms all requests have complete windows.

10. executive summary

Turn 20. The executive summary brings together the result, target, comparison, exceptions, and limits in a concise repeatable account.

Return to the conversation, page 35

03 / Answers and explanations / continued

F / Accuracy and scope

F1. What is the approximate relative increase from 90% to 96%?

D. 6.67%

The six-point difference divided by the ninety-percent reference rate is approximately 6.67%.

Return to the exercise, page 39

F2. What cannot be concluded from the supplied totals?

B. The specific cause of the improvement

The briefing supplies outcomes and comparable definitions but no causal investigation evidence.

Return to the exercise, page 39

F / Guided speaking

1. 90

Analyst: This month is 90 percent.

Turn 1. One hundred eighty divided by two hundred equals ninety percent.

2. 85

Manager: Last month was 85 percent.

Turn 2. One hundred seventy divided by two hundred equals eighty-five percent.

3. five

Analyst: The month-on-month increase is five percentage points.

Turn 3. Ninety percent minus eighty-five percent is five percentage points.

4. two

Manager: This month exceeds target by two percentage points.

Turn 4. Ninety percent minus the eighty-eight-percent target equals two percentage points.

Return to the conversation, page 40

03 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Four hundred eighty needs a denominator and a target". Elena: "Can I call it a sustained improvement, or do two months only support the month-on-month comparison? I want to recognize the result accurately." Jonas: [missing response] Elena: "Fair. We can acknowledge the result without attributing it to a particular staffing change that has not been evaluated. The definition should stay visible too." Which response belongs in the gap?

D. We can report the month-on-month improvement. Two months do not establish a sustained trend, and these totals do not identify the cause. Recognition does not require a stronger statistical story.

A Not here. This line belongs to turn 12, after Elena says: "I also want to avoid the green arrow hiding the twenty late requests. Meeting the overall threshold does not mean every customer received the target service." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Elena says: "August had four hundred fifty timely requests out of five hundred. The same definition and exclusions apply, so that earlier rate was ninety percent." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Elena says: "There were five hundred eligible requests in September, and the other twenty finished late. All requests have had enough time for us to observe their outcomes." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Jonas's response in turn 14. It connects Elena's preceding point with Elena's follow-up. Month-on-month describes the immediate September-versus-August comparison without claiming a longer established pattern.

Return to the exercise, page 40

04 / Answers and explanations

A / Read the situation

A1. What is the attendee renewal rate?

D. 90%

Ninety renewals divided by one hundred eligible attendees equals ninety percent.

Return to the exercise, page 41

A2. What is the non-attendee renewal rate?

C. 80%

One hundred sixty renewals divided by two hundred eligible non-attendees equals eighty percent.

Return to the exercise, page 41

A3. Why is the causal effect not established?

B. Attendance was voluntary, and unmeasured baseline differences may matter.

Voluntary selection and possible unmeasured differences prevent the observed comparison from isolating the training effect.

Return to the exercise, page 41

D / Response choices

D1. Which headline is supported?

B. Higher observed renewal among voluntary training attendees

The supported headline describes the observed association without asserting an unestablished causal result.

Return to the exercise, page 45

D2. What is the relative observed difference compared with 80%?

A. 12.5%

The ten-point gap divided by the eighty-percent reference rate equals 12.5%.

Return to the exercise, page 45

04 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Distortion from factors related to both exposure and outcome. Which term fits?

B. confounding

A Not here. renewal: Continuation of an eligible customer contract or relationship. The usual combination is "measure customer renewal".

B Correct. confounding: Distortion from factors related to both exposure and outcome. The usual combination is "assess potential confounding".

C Not here. unmeasured factor: A relevant characteristic not recorded in the data. The usual combination is "disclose unmeasured factors".

D Not here. engagement: Participation or involvement under a specified definition. The usual combination is "define customer engagement".

Return to the exercise, page 45

D4. Rebuild the exchange from "Training attendees renewed more often". Marcus: "Then the ten-point difference could reflect training, pre-existing differences, or both. We should not assume the entire gap was created by the sessions." Priya: [missing response] Marcus: "I do not want the caveat to erase the positive finding. The team has real evidence that attendees renewed more often in this group." Which response belongs in the gap?

B. Exactly. The causal effect asks what the intervention changes compared with what would otherwise have happened. This comparison does not isolate that quantity simply because the rates are clearly different.

A Not here. This line belongs to turn 14, after Marcus says: "The business question is whether offering or requiring training would improve renewal for eligible customers. That is different from describing people who already chose to attend." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Priya's response in turn 6. It connects Marcus's preceding point with Marcus's follow-up. The causal effect concerns what training itself changes, which the voluntary comparison does not isolate.

C Not here. This line belongs to turn 12, after Marcus says: "For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Marcus says: "Sales will ask whether you are saying motivation caused the whole gap. We did not measure it, so that is not our conclusion either, is it?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 45

04 / Answers and explanations / continued

E / Conversation

1. association

Turn 2. The association is the observed difference between voluntary attendance groups, not an isolated effect of training.

2. self-selection

Turn 4. Self-selection means customers chose attendance, which can create relevant differences between the compared groups.

3. causal effect

Turn 6. The causal effect concerns what training itself changes, which the voluntary comparison does not isolate.

4. Confounding

Turn 8. Confounding is a possible distortion from factors related to both attendance and renewal, not proof that the observed association is meaningless.

5. unmeasured factor

Turn 10. Motivation is an unmeasured factor that could matter, but the case does not establish its size or explanatory contribution.

6. relative lift

Turn 12. Relative lift divides the ten-point observed difference by the eighty-percent comparison rate, giving 12.5% without establishing causation.

7. counterfactual

Turn 14. The counterfactual concerns what the relevant customers would have done under a different training exposure, which is not directly observed here.

8. adjustment

Turn 16. Adjustment can account for specified measured variables but does not automatically remove all unmeasured differences or prove a causal effect.

9. generalizability

Turn 18. Generalizability asks whether a result applies beyond the studied voluntary groups to the broader proposed audience.

04 / Answers and explanations / continued

10. evaluation design

Turn 20. The evaluation design specifies how the next assessment will address the causal question more credibly than the current voluntary comparison.

Return to the conversation, page 47

F / Accuracy and scope

F1. Which statement about motivation is accurate?

D. It is an unmeasured possible source of group differences.

The briefing identifies motivation as a possible difference but supplies no measurement or randomization.

Return to the exercise, page 51

F2. What should a revised recommendation do?

A. Preserve the association and seek a reviewed design for a causal evaluation.

A better evaluation can address the causal question while the current association remains honestly reported.

Return to the exercise, page 51

F / Guided speaking

1. 90

Analyst: Attendee renewal is 90 percent.

Turn 1. Forty-five divided by fifty equals ninety percent for attendees.

2. 80

Lead: Non-attendee renewal is 80 percent.

Turn 2. Eighty divided by one hundred equals eighty percent for non-attendees.

3. ten

Analyst: The observed gap is ten percentage points.

Turn 3. Ninety percent minus eighty percent equals ten percentage points.

4. unestablished

Lead: The causal effect remains unestablished.

Turn 4. Voluntary attendance and unmeasured baseline differences prevent this comparison from isolating causation.

Return to the conversation, page 52

04 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Training attendees renewed more often". Marcus: "The business question is whether offering or requiring training would improve renewal for eligible customers. That is different from describing people who already chose to attend." Priya: [missing response] Marcus: "We may be able to add customer size and tenure. Would adjusting for those settle the concern, or would important differences still be missing?" Which response belongs in the gap?

A. Yes. The counterfactual question is what comparable customers would do without the intervention. We need a reviewed evaluation approach suited to that decision, not just a more persuasive headline.

A Correct. This is Priya's response in turn 14. It connects Marcus's preceding point with Marcus's follow-up. The counterfactual concerns what the relevant customers would have done under a different training exposure, which is not directly observed here.

B Not here. This line belongs to turn 12, after Marcus says: "For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 10, after Marcus says: "Sales will ask whether you are saying motivation caused the whole gap. We did not measure it, so that is not our conclusion either, is it?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 6, after Marcus says: "Then the ten-point difference could reflect training, pre-existing differences, or both. We should not assume the entire gap was created by the sessions." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 52

05 / Answers and explanations

A / Read the situation

A1. How many rows result from the supplied join?

D. Three

O1 matches two payment rows and O2 matches one, producing three joined rows.

Return to the exercise, page 56

A2. What is the actual total order value?

B. $200

Two distinct orders worth one hundred dollars each total two hundred dollars.

Return to the exercise, page 56

A3. Why does SUM(DISTINCT order_amount) give $100 here?

D. Both legitimate orders have the same amount, so distinct values collapse them.

Distinct amounts are not distinct orders when different orders can share the same value.

Return to the exercise, page 56

D / Response choices

D1. Which operation explains the $300 result?

D. Summing order amounts after O1 is repeated for two payments

The joined order measure repeats O1's one-hundred-dollar value and inflates the aggregate.

Return to the exercise, page 60

D2. Why is summing distinct amounts not a valid repair?

B. It merges different orders that happen to have equal values.

A value-level distinct operation does not deduplicate business entities by their keys.

Return to the exercise, page 60

05 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A column or set of columns constrained to reference a key in a table. Which term fits?

C. foreign key

A Not here. fact table: A table storing business events or measurements at a defined grain. The usual combination is "define the fact-table grain".

B Not here. aggregation: Combining records into summary values. The usual combination is "control aggregation grain".

C Correct. foreign key: A column or set of columns constrained to reference a key in a table. The usual combination is "check foreign-key relationships".

D Not here. inner join: A join returning matching combinations from both sides. The usual combination is "inspect inner-join matches".

Return to the exercise, page 60

D4. Rebuild the exchange from "The join adds rows, not order value". Owen: "I summed the order amount after the join. Each resulting row displays one hundred, so the three displayed amounts add to three hundred." Lina: [missing response] Owen: "SUM DISTINCT brought it down to one hundred. That looks worse: we have two separate orders worth one hundred each. Have I removed the wrong duplicate?" Which response belongs in the gap?

A. That is join fanout. The payment rows are legitimate, but the order-level measure is repeated. We need to preserve its meaning instead of treating each joined occurrence as new value.

A Correct. This is Lina's response in turn 6. It connects Owen's preceding point with Owen's follow-up. Fanout multiplies the order row through payment matches, so summing the repeated order measure inflates its total.

B Not here. This line belongs to turn 18, after Owen says: "I will check both the two-order count and the two-hundred-dollar total. Could those totals look right even if one missing order offsets a duplicate?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Owen says: "Both orders in this extract have payments. But the full report must still include an order without a payment if it belongs in the reporting population." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Owen says: "The correct order total is two hundred. The payment amounts also total two hundred in this small dataset, but that does not make the measures identical." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 60

05 / Answers and explanations / continued

E / Conversation

1. row grain

Turn 2. Row grain describes what each row represents, which changes when an order is matched to multiple payment rows.

2. one-to-many relationship

Turn 4. The one-to-many relationship lets one order match multiple payment records, producing more than one output row for O1.

3. fanout

Turn 6. Fanout multiplies the order row through payment matches, so summing the repeated order measure inflates its total.

4. DISTINCT

Turn 8. DISTINCT on the amount removes equal numeric values, not repeated order identities, so both legitimate hundred-dollar orders collapse.

5. aggregation

Turn 10. Aggregation combines rows at a chosen grain; the measure and that grain must match to avoid repeated order values.

6. pre-aggregation

Turn 12. Pre-aggregation summarizes payments before the join so each order matches at most one summarized payment row.

7. join key

Turn 14. The join key is order ID, which identifies the business relationship rather than a coincidentally equal amount.

8. left join

Turn 16. A left join can retain orders without a matching payment summary, but its retention behavior does not itself prevent fanout.

9. reconciliation total

Turn 18. The reconciliation total is the two-hundred-dollar source order value used to verify that the transformed report preserves the intended measure.

05 / Answers and explanations / continued

10. uniqueness test

Turn 20. The uniqueness test checks one row per order ID in the summarized data so the corrected relationship can be maintained.

Return to the conversation, page 62

F / Accuracy and scope

F1. Which proposed repair fits this report?

B. Summarize payments by order ID, then join at one row per order and verify totals.

Pre-aggregation by order ID preserves business identity and aligns the grain. Distinct amounts merge legitimate orders, amount joins use the wrong key, and a left join alone does not prevent multiple matches.

Return to the exercise, page 65

F2. Which statement about a left join is accurate?

A. It can preserve unmatched orders but still duplicate an order with multiple matches.

Left retention and match multiplicity are separate properties of the join.

Return to the exercise, page 65

F / Guided speaking

1. 160

Engineer: The correct order total is $160.

Turn 1. Two distinct eighty-dollar orders total one hundred sixty dollars.

2. 240

Analyst: The joined sum of order amounts is $240.

Turn 2. Three joined rows each repeat an eighty-dollar order amount.

3. 80

Engineer: The sum of distinct amount values is $80.

Turn 3. The equal eighty-dollar values collapse to one distinct numeric amount.

4. identity

Analyst: The repair must preserve order identity.

Turn 4. Correct deduplication depends on distinct order identity, not distinct amount values.

Return to the conversation, page 66

05 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "The join adds rows, not order value". Owen: "The grouping field should be order ID, not amount. Grouping by amount would combine the two orders again because they both have the same value." Lina: [missing response] Owen: "Both orders in this extract have payments. But the full report must still include an order without a payment if it belongs in the reporting population." Which response belongs in the gap?

A. Correct. The join key must preserve the intended relationship. Test the summary key and check that the matching logic uses the appropriate identifier on both sides.

A Correct. This is Lina's response in turn 14. It connects Owen's preceding point with Owen's follow-up. The join key is order ID, which identifies the business relationship rather than a coincidentally equal amount.

B Not here. This line belongs to turn 4, after Owen says: "O1 is one hundred dollars with payments of sixty and forty. O2 is another one-hundred-dollar order with a single one-hundred-dollar payment." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Owen says: "SUM DISTINCT brought it down to one hundred. That looks worse: we have two separate orders worth one hundred each. Have I removed the wrong duplicate?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Owen says: "The report jumped from two hundred dollars of order value to three hundred after I added payment details. I thought the join would only add columns." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 66

06 / Answers and explanations

A / Read the situation

A1. What is the point estimate?

C. +1.2 percentage points

The supplied central effect estimate is positive one point two percentage points.

Return to the exercise, page 67

A2. Did the lower-bound rule pass?

A. No; -0.4 does not exceed +1.0.

The rule concerns the interval lower bound, not the positive point estimate.

Return to the exercise, page 67

A3. What stage has the test reached?

B. Its planned endpoint

The briefing explicitly says the planned endpoint has been reached without new authorization.

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D / Response choices

D1. Which slide is most faithful to the readout?

D. +1.2 points; 95% interval -0.4 to +2.8; guardrails pass; rollout criterion unmet

The correct slide retains percentage-point units, the negative lower bound, and both conditions of the rollout rule. The alternatives change the decision, scale, or sign.

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D2. Why does the +1.2 estimate not satisfy the rule?

C. The rule uses the interval lower bound, which is -0.4.

The pre-agreed rule explicitly concerns the lower endpoint exceeding one point.

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06 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: An interval from a method with specified long-run coverage. Which term fits?

A. confidence interval

A Correct. confidence interval: An interval from a method with specified long-run coverage. The usual combination is "interpret a confidence interval".

B Not here. minimum detectable effect: An effect size used in the study's power planning. The usual combination is "specify the minimum detectable effect".

C Not here. no-change value: The parameter value representing no effect on the stated scale. The usual combination is "identify the no-change value".

D Not here. statistical significance: A result meeting a specified statistical testing criterion. The usual combination is "qualify statistical significance".

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D4. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "Remind me of the actual rollout rule. I remember that we set a one-percentage-point threshold and also required the guardrails to pass." Theo: [missing response] Hana: "The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement." Which response belongs in the gap?

C. The lower bound must exceed plus one point zero, with guardrails passing. Our lower endpoint is minus zero point four, so that condition was not met.

A Not here. This line belongs to turn 4, after Hana says: "That range includes a small negative effect and zero as well as useful positive outcomes. The current slide hides those possibilities by showing only the center." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 20, after Hana says: "My revised readout will show the center, both bounds, passing guardrails, and the unmet rollout condition. No launch approval or no-effect conclusion will be implied." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Theo's response in turn 6. It connects Hana's preceding point with Hana's follow-up. The lower bound is minus 0.4 points, which fails the requirement that it exceed plus 1.0 point.

D Not here. This line belongs to turn 18, after Hana says: "I will also keep the units visible. Plus one point two means percentage points, not a one-point-two-percent relative increase from an unstated baseline." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 71

E / Conversation

1. point estimate

Turn 2. The point estimate is the single plus-1.2-point estimate, which must be interpreted with its uncertainty and the decision rule.

06 / Answers and explanations / continued

2. confidence interval

Turn 4. The confidence interval communicates uncertainty under its statistical method instead of treating the estimate as a guaranteed effect.

3. lower bound

Turn 6. The lower bound is minus 0.4 points, which fails the requirement that it exceed plus 1.0 point.

4. decision rule

Turn 8. The decision rule requires both the lower-bound condition and passing guardrails, so one passing condition cannot replace the other.

5. no-change value

Turn 10. Zero is the no-change value on the difference scale, but its presence in the interval does not prove the effect is exactly zero.

6. inconclusive result

Turn 12. The result is inconclusive for the desired positive-effect conclusion, while the supplied rollout rule has a definite unmet condition.

7. planned endpoint

Turn 14. The planned endpoint has already been reached, so an opportunistic extension would change the agreed analysis behavior.

8. sequential analysis

Turn 16. Sequential analysis is designed for planned repeated decisions; it cannot be assumed to justify an unapproved extension of this completed test.

9. practical threshold

Turn 18. The practical threshold is the one-point business-relevant boundary used in the agreed lower-bound decision condition.

10. practical significance

Turn 20. Practical significance concerns whether an effect matters for the business, which is not established merely by a positive central estimate.

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06 / Answers and explanations / continued

F / Accuracy and scope

F1. What would be an unsupported next step?

D. Extend the finished test repeatedly until the original method produces a favorable result.

Opportunistic extension changes the analysis behavior without the approved method needed for valid inference.

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F2. Which statement about zero is correct?

D. Its inclusion does not prove either exact zero effect or equivalence.

An interval containing zero permits uncertainty and does not establish an equivalence conclusion.

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F / Guided speaking

1. 0.8

Analyst: The point estimate is plus 0.8 points.

Turn 1. The supplied central estimate is positive zero point eight percentage points.

2. 0.2

Director: The lower bound is minus 0.2 points.

Turn 2. The interval begins at negative zero point two percentage points.

3. 0.5

Analyst: The required boundary is plus 0.5 points.

Turn 3. The stated lower-bound rule uses positive zero point five as its threshold.

4. unmet

Director: The rollout condition is unmet.

Turn 4. The negative lower bound does not exceed the required positive threshold.

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06 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "Could we extend it until the lower bound clears one point? Or would that change the rule after we have already seen the result?" Theo: [missing response] Hana: "A new study might still be justified, but it would need a clear design and rationale rather than an attempt to rescue the original headline." Which response belongs in the gap?

B. We reached the planned endpoint. Continuing until the result looks favorable changes the statistical behavior. There is no authorized extension or approved repeated-decision method in the supplied plan.

A Not here. This line belongs to turn 10, after Hana says: "Then should the slide say no effect? Zero is inside the interval, but so are positive effects. I do not want to replace one overclaim with another." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Theo's response in turn 14. It connects Hana's preceding point with Hana's follow-up. The planned endpoint has already been reached, so an opportunistic extension would change the agreed analysis behavior.

C Not here. This line belongs to turn 8, after Hana says: "The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 16, after Hana says: "A new study might still be justified, but it would need a clear design and rationale rather than an attempt to rescue the original headline." It addresses that point, not the missing turn in the supplied exchange.

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07 / Answers and explanations

A / Read the situation

A1. Which fields directly support the supplied analysis?

B. Week, tier, eligible-case count, and late-case count

The question concerns weekly tier-level rates, which the four specified aggregate fields can support.

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A2. What is the current approval status?

A. Pending

The briefing explicitly leaves owner approval and review incomplete.

Return to the exercise, page 81

A3. What does replacing names with stable IDs establish?

B. It does not automatically establish anonymity.

Stable identifiers may still allow linkage, so name removal alone does not establish anonymity.

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D / Response choices

D1. Which request best matches the supplied need?

A. Reviewed weekly tier-level counts in the approved workspace, with pending approvals and 30-day access

The correct request supplies the needed measures while preserving the stated handling conditions.

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D2. Why include eligible-case count?

A. It provides the denominator for each rate.

A late-case count alone cannot show the proportion of eligible cases affected.

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07 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Restricting data use to defined appropriate purposes. Which term fits?

D. purpose limitation

A Not here. access scope: The specific data and actions a permission covers. The usual combination is "limit access scope".

B Not here. processing purpose: The defined reason for collecting or using data. The usual combination is "state the processing purpose".

C Not here. pseudonymization: Replacing identifying details while retaining possible linkage. The usual combination is "assess pseudonymization safeguards".

D Correct. purpose limitation: Restricting data use to defined appropriate purposes. The usual combination is "document the analysis purpose".

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D4. Rebuild the exchange from "The analysis needs weekly counts, not customer names". Sara: "An aggregate table would work for this comparison. I do not need individual rows if the counts use the agreed definitions and can be traced to their source." Malik: [missing response] Sara: "So a small tier in one week should not automatically be published just because its row contains counts rather than names. The audience and context still matter." Which response belongs in the gap?

C. That aggregation is a useful alternative, subject to review. We still need to check the output, especially where a group is small enough to raise identification or disclosure concerns.

A Not here. This line belongs to turn 4, after Sara says: "Those extra fields do not enter the rate. I need late cases divided by eligible cases within each week and tier." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Sara says: "So a small tier in one week should not automatically be published just because its row contains counts rather than names. The audience and context still matter." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Malik's response in turn 6. It connects Sara's preceding point with Sara's follow-up. Aggregation supplies group-level counts without automatically resolving disclosure risks or approval requirements.

D Not here. This line belongs to turn 16, after Sara says: "I will also say who needs the results. The service manager needs the reviewed weekly comparison, but that does not imply that anyone may receive the underlying data." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 85

07 / Answers and explanations / continued

E / Conversation

1. processing purpose

Turn 2. The processing purpose is the weekly tier-level comparison that determines which fields the analysis needs.

2. data minimization

Turn 4. Data minimization limits the request to sufficient relevant information rather than collecting unrelated personal fields for possible future use.

3. aggregation

Turn 6. Aggregation supplies group-level counts without automatically resolving disclosure risks or approval requirements.

4. disclosure review

Turn 8. Disclosure review assesses sharing risks, including small groups and the proposed audience.

5. pseudonymization

Turn 10. Pseudonymization can replace direct names while preserving possible linkage; it does not automatically establish anonymity or remove controls.

6. access scope

Turn 12. Access scope defines the permitted data and actions, including the approved workspace rather than an assumed unrestricted export.

7. Access expiry

Turn 14. Access expiry ends permission; separate approved rules govern how long the data is retained.

8. permitted audience

Turn 16. The permitted audience identifies who may receive the approved result and prevents internal approval from becoming an unrestricted-sharing assumption.

9. secondary use

Turn 18. Secondary use is a different use beyond the original purpose and may require a new assessment instead of inheriting approval automatically.

10. data lineage

Turn 20. Data lineage records the source and transformations supporting the aggregate counts so their meaning and quality remain traceable.

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07 / Answers and explanations / continued

F / Accuracy and scope

F1. Which statement about aggregation is accurate?

A. It can reduce detail but still needs the stated disclosure review.

The case explicitly requires review, including small-group risks, despite the aggregate format.

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F2. What does 30-day access specify?

C. The proposed duration of permission in this case

The briefing defines a local access proposal, not a general retention law or completed approval.

Return to the exercise, page 91

F / Guided speaking

1. count

Analyst: The rate denominator is the eligible-order count.

Turn 1. Eligible-order count provides the base for the complaint rate.

2. unnecessary

Steward: Phone numbers are unnecessary for this question.

Turn 2. The briefing explicitly says phone numbers do not support the stated aggregate analysis.

3. 14

Analyst: The proposed access period is 14 days.

Turn 3. The supplied local proposal specifies fourteen days of access.

4. pending

Steward: Approval remains pending.

Turn 4. The briefing says access approval has not yet been granted.

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07 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "The analysis needs weekly counts, not customer names". Sara: "Does thirty-day access also mean deleting the data on day thirty? I need to distinguish the permission expiry from the actual retention rule." Malik: [missing response] Sara: "I will also say who needs the results. The service manager needs the reviewed weekly comparison, but that does not imply that anyone may receive the underlying data." Which response belongs in the gap?

C. Exactly. Access expiry and data retention are separate questions. The actual retention and deletion arrangements need their own approved basis rather than being inferred from this access duration.

A Not here. This line belongs to turn 6, after Sara says: "An aggregate table would work for this comparison. I do not need individual rows if the counts use the agreed definitions and can be traced to their source." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Sara says: "So a small tier in one week should not automatically be published just because its row contains counts rather than names. The audience and context still matter." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Malik's response in turn 14. It connects Sara's preceding point with Sara's follow-up. Access expiry ends permission; separate approved rules govern how long the data is retained.

D Not here. This line belongs to turn 20, after Sara says: "My request will list the four fields, rate, workspace, thirty-day access, audience, and required reviews. Approval remains pending." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 92

08 / Answers and explanations

A / Read the situation

A1. How did the issue-Q ticket rate change?

B. From 2% to 8%, a six-percentage-point rise

Twenty and eighty divided by one thousand give two and eight percent, respectively.

Return to the exercise, page 94

A2. What does the mobile count establish?

B. Seventy of the eighty later issue-Q tickets are mobile-related.

The supplied count describes the composition of issue-Q tickets, not a mobile population failure rate.

Return to the exercise, page 94

A3. What authorization is requested?

A. A two-day investigation with a subsequent readout

The briefing defines a bounded investigation rather than a fix, cause conclusion, or broader permission.

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D / Response choices

D1. Which opening best uses the five-minute meeting?

A. Approve a two-day investigation of Q; its ticket rate rose from 2% to 8%.

The opening connects a specific decision to the strongest supplied observation.

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D2. What is the relative increase from 2% to 8%?

C. 300%

The six-point increase divided by the two-percent original rate equals three hundred percent.

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08 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The main evidence statement in a short briefing. Which term fits?

A. headline finding

A Correct. headline finding: The main evidence statement in a short briefing. The usual combination is "sharpen the headline finding".

B Not here. base rate: The underlying frequency relative to an eligible population. The usual combination is "check the base rate".

C Not here. investigation owner: The role accountable for the defined inquiry. The usual combination is "assign the investigation owner".

D Not here. priority criterion: A stated basis for ordering work. The usual combination is "apply the priority criterion".

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D4. Rebuild the exchange from "Five minutes to authorize the right investigation". Claudia: "That is six percentage points, and four times the earlier rate. We should not call it six-percent growth or accidentally present ticket counts as unique customers." Oren: [missing response] Claudia: "Seventy of the eighty are mobile-related. Can I say mobile fails more often than desktop, or are we missing the denominators for that comparison?" Which response belongs in the gap?

A. Correct. The headline finding is a rise in the issue's ticket rate, not proof of how many distinct people were affected or why the issue increased.

A Correct. This is Oren's response in turn 6. It connects Claudia's preceding point with Claudia's follow-up. The headline finding states the observed ticket-rate movement with its correct units rather than implying distinct users or a causal explanation.

B Not here. This line belongs to turn 18, after Claudia says: "Does approving this request also allow the analyst to change production if a likely cause appears? I want that distinction written clearly before I decide." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 10, after Claudia says: "But the concentration still gives the investigator a useful place to start. We can inspect the mobile paths without overstating what seventy out of eighty means." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Claudia says: "Then the decision is whether to allocate two days to a documented service issue, with the mobile concentration as a lead and the cause still open." It addresses that point, not the missing turn in the supplied exchange.

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E / Conversation

1. recommendation

Turn 2. The recommendation is the proposed two-day investigation tied to the documented rise in issue-Q ticket frequency.

08 / Answers and explanations / continued

2. supporting evidence

Turn 4. The supporting evidence supplies twenty and eighty issue tickets with the same one-thousand-ticket denominator in each week.

3. headline finding

Turn 6. The headline finding states the observed ticket-rate movement with its correct units rather than implying distinct users or a causal explanation.

4. segment concentration

Turn 8. Segment concentration describes where the issue tickets are concentrated, not the failure frequency within all mobile activity.

5. denominator gap

Turn 10. The denominator gap is the missing mobile base required to turn the issue composition into a mobile-specific rate.

6. causal hypothesis

Turn 12. The causal hypothesis is the possible release explanation, which needs evaluation rather than being asserted from timing alone.

7. investigation owner

Turn 14. The investigation owner is the service analyst responsible for the defined inquiry and subsequent evidence readout.

8. time box

Turn 16. The time box limits the investigation duration and creates a review point without guaranteeing a causal conclusion or completed fix.

9. action boundary

Turn 18. The action boundary restricts the approval to investigation and readout, leaving production changes subject to their separate process.

10. decision ask

Turn 20. The decision ask is the precise approval or rejection of the defined two-day inquiry, not endorsement of an unproven cause.

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08 / Answers and explanations / continued

F / Accuracy and scope

F1. Why is 70/80 not a mobile failure rate?

D. Its denominator is issue-Q tickets, not all relevant mobile activity.

The supplied denominator describes issue composition and cannot substitute for the missing mobile base.

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F2. What does approval of this request authorize?

D. The defined two-day investigation and readout

The stated action boundary limits approval to the investigation, not implementation or causal claims.

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F / Guided speaking

1. six

Analyst: The later issue rate is six percent.

Turn 1. Thirty divided by five hundred equals six percent of all tickets.

2. four

Director: The increase is four percentage points.

Turn 2. The earlier rate is two percent, so the increase is four points.

3. 80

Analyst: Mobile represents 80 percent of later issue tickets.

Turn 3. Twenty-four divided by thirty equals eighty percent of issue tickets, not mobile users.

4. investigation

Director: Approval would authorize an investigation.

Turn 4. The supplied request is for investigation only, not a production modification.

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08 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Five minutes to authorize the right investigation". Claudia: "Who would lead the work? I want a named role and a defined output, not an open request for somebody to analyze the problem someday." Oren: [missing response] Claudia: "Two days for an evidence readout, not a guaranteed root cause. Please make that explicit in the invitation before people start expecting a fix." Which response belongs in the gap?

B. The service analyst is the proposed investigation owner. The output is a readout of evidence, remaining uncertainty, and a recommended next step, rather than an unsupported promise of a fix.

A Not here. This line belongs to turn 4, after Claudia says: "Give me the actual counts as well. I want to know whether the percentage changed because the ticket base became much smaller." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Oren's response in turn 14. It connects Claudia's preceding point with Claudia's follow-up. The investigation owner is the service analyst responsible for the defined inquiry and subsequent evidence readout.

C Not here. This line belongs to turn 16, after Claudia says: "Two days for an evidence readout, not a guaranteed root cause. Please make that explicit in the invitation before people start expecting a fix." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Claudia says: "Seventy of the eighty are mobile-related. Can I say mobile fails more often than desktop, or are we missing the denominators for that comparison?" It addresses that point, not the missing turn in the supplied exchange.

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Review 1 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Rules specifying exactly how a measure is calculated. Which term fits?

C. metric definition

A Not here. transaction-based activity: Activity defined by a qualifying transaction. The usual combination is "report transaction-based activity".

B Not here. denominator: The total used as the base of a rate. The usual combination is "state the denominator".

C Correct. metric definition: Rules specifying exactly how a measure is calculated. The usual combination is "agree a metric definition".

D Not here. internal account: An account belonging to the organization's own personnel or systems. The usual combination is "exclude internal accounts".

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2. A colleague needs the term for this meaning: Absence of unintended duplicate records. Which term fits?

A. uniqueness

A Correct. uniqueness: Absence of unintended duplicate records. The usual combination is "test record uniqueness".

B Not here. data quality: Fitness of data for its intended use. The usual combination is "assess data quality".

C Not here. coverage window: The period or population represented by available data. The usual combination is "align coverage windows".

D Not here. upstream source: An earlier system supplying data to a pipeline. The usual combination is "inspect upstream sources".

Return to the exercise, page 27

Review 1 / Explanations / continued

3. Rebuild the exchange from "Two active-user totals, two different events". Rafael: "I have seventy-five percent on the slide: nine hundred divided by twelve hundred. Is that answering the eligible-user question, or have I changed the base?" Mei: [missing response] Rafael: "Both rates tell us something, but forty-five percent is the answer to the eligible-user question. Seventy-five percent describes the logged-in group instead." Which response belongs in the gap?

C. That uses a different denominator. Nine hundred divided by two thousand eligible users is forty-five percent. Nine hundred divided by twelve hundred logged-in users is seventy-five percent.

A Not here. This line belongs to turn 18, after Rafael says: "Who signs off if the transaction event changes next month? I do not want the same label to hide a new definition." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 20, after Rafael says: "My summary will be nine hundred transaction-active users, forty-five percent of two thousand eligible users, September UTC. The login measure remains separately labeled." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Mei's response in turn 10. It connects Rafael's preceding point with Rafael's follow-up. The denominator defines the rate's base; eligible users and logged-in users answer different percentage questions.

D Not here. This line belongs to turn 6, after Rafael says: "The window is the September calendar month in UTC for both, and they use the same two thousand eligible users. That part is aligned." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 27

Review 1 / Explanations / continued

4. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "When the delivery arrives, who checks that it is complete? I do not want the warning removed just because the refresh turns green." Asha: [missing response] Tomas: "I will take the Monday-only view to the meeting, explain the Tuesday refund gap, and avoid promising a two-day net result before it is verified." Which response belongs in the gap?

B. Validate the backfill and document the checks. A green refresh icon should not be our only evidence that the previously incomplete reporting period is now fit for use.

A Not here. This line belongs to turn 2, after Tomas says: "The dashboard says refreshed this morning. Can I use the Monday-Tuesday net-order-value number in the operations meeting as our complete two-day result?" It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Asha's response in turn 18. It connects Tomas's preceding point with Tomas's follow-up. The backfill loads previously missing records, which need verification before the incomplete-status notice can be removed.

C Not here. This line belongs to turn 12, after Tomas says: "The note should be near the result, not somewhere people will miss it. I will explain that Tuesday orders alone do not settle the net figure." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Tomas says: "I had assumed that a successful refresh meant every table contained the latest business activity. Which boundary should I read instead for the refund data?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 27

Review 2 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A specified desired level of a measure. Which term fits?

A. performance target

A Correct. performance target: A specified desired level of a measure. The usual combination is "compare with the performance target".

B Not here. annotation: Explanatory text attached to a chart or figure. The usual combination is "add a relevant annotation".

C Not here. observation window: The period available to observe the relevant outcome. The usual combination is "complete the observation window".

D Not here. relative change: Change divided by the original reference value. The usual combination is "calculate relative change".

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2. A colleague needs the term for this meaning: A person's underlying inclination to act or engage. Which term fits?

C. motivation

A Not here. association: An observed relationship between variables or groups. The usual combination is "report an association".

B Not here. relative lift: A difference expressed relative to a comparison value. The usual combination is "qualify relative lift".

C Correct. motivation: A person's underlying inclination to act or engage. The usual combination is "assess baseline motivation".

D Not here. comparison group: The group used as a reference for the outcome. The usual combination is "select a comparison group".

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Review 2 / Explanations / continued

3. Rebuild the exchange from "Refreshed today, complete only through Monday". Tomas: "I still need a reliable number for the meeting. Could we show Monday by itself, since both sources are verified complete through that day?" Asha: [missing response] Tomas: "The note should be near the result, not somewhere people will miss it. I will explain that Tuesday orders alone do not settle the net figure." Which response belongs in the gap?

A. Yes. That respects completeness for the shared period in this case. We can publish the Monday-only comparison and make clear that the two-day figure remains incomplete.

A Correct. This is Asha's response in turn 10. It connects Tomas's preceding point with Tomas's follow-up. Completeness concerns the required records for the stated period; the briefing verifies both sources for Monday.

B Not here. This line belongs to turn 6, after Tomas says: "Then the two-day net figure has Tuesday orders but not the corresponding Tuesday refunds. It could look better simply because that input is incomplete." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Tomas says: "I had assumed that a successful refresh meant every table contained the latest business activity. Which boundary should I read instead for the refund data?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Tomas says: "The dashboard says refreshed this morning. Can I use the Monday-Tuesday net-order-value number in the operations meeting as our complete two-day result?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 54

Review 2 / Explanations / continued

4. Rebuild the exchange from "Training attendees renewed more often". Marcus: "And a mandatory program could reach people unlike the voluntary attendees. Even a stronger estimate in this group might not transfer unchanged to everyone." Priya: [missing response] Marcus: "I will revise the slide to higher observed renewal among attendees, show both counts and rates, and recommend a better evaluation before claiming a guaranteed benefit." Which response belongs in the gap?

A. That is a generalizability issue as well as a causal one. The population, intervention, and implementation matter when translating evidence into a wider operational recommendation.

A Correct. This is Priya's response in turn 18. It connects Marcus's preceding point with Marcus's follow-up. Generalizability asks whether a result applies beyond the studied voluntary groups to the broader proposed audience.

B Not here. This line belongs to turn 4, after Marcus says: "Both groups had the same renewal window and eligibility rules. That makes the comparison cleaner, even though attendance was voluntary rather than assigned." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Marcus says: "Training attendees renewed at ninety percent, versus eighty percent for non-attendees. Can we say the sessions improved renewal by ten points?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Marcus says: "I do not want the caveat to erase the positive finding. The team has real evidence that attendees renewed more often in this group." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 54

Review 3 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A non-null column or set of columns uniquely identifying each row. Which term fits?

A. primary key

A Correct. primary key: A non-null column or set of columns uniquely identifying each row. The usual combination is "validate the primary key".

B Not here. COUNT: An aggregate that counts rows or non-null expressions. The usual combination is "distinguish COUNT variants".

C Not here. fanout: Multiplication of rows caused by multiple join matches. The usual combination is "detect join fanout".

D Not here. data model: A representation of entities, relationships, and measures. The usual combination is "review the data model".

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2. A colleague needs the term for this meaning: The magnitude of a difference or relationship. Which term fits?

D. effect size

A Not here. primary outcome: The main measure used to assess the intervention. The usual combination is "specify the primary outcome".

B Not here. randomized experiment: A study allocating exposure through a random process. The usual combination is "interpret a randomized experiment".

C Not here. sampling variability: Variation in estimates across possible samples. The usual combination is "account for sampling variability".

D Correct. effect size: The magnitude of a difference or relationship. The usual combination is "assess effect size".

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Review 3 / Explanations / continued

3. Rebuild the exchange from "Training attendees renewed more often". Marcus: "Sales will ask whether you are saying motivation caused the whole gap. We did not measure it, so that is not our conclusion either, is it?" Priya: [missing response] Marcus: "For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?" Which response belongs in the gap?

D. Right. It is an unmeasured factor that may matter, not a proven explanation of every percentage point. We should avoid replacing one unsupported causal story with another equally unsupported story.

A Not here. This line belongs to turn 12, after Marcus says: "For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 14, after Marcus says: "The business question is whether offering or requiring training would improve renewal for eligible customers. That is different from describing people who already chose to attend." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 6, after Marcus says: "Then the ten-point difference could reflect training, pre-existing differences, or both. We should not assume the entire gap was created by the sessions." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Priya's response in turn 10. It connects Marcus's preceding point with Marcus's follow-up. Motivation is an unmeasured factor that could matter, but the case does not establish its size or explanatory contribution.

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Review 3 / Explanations / continued

4. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "I will also keep the units visible. Plus one point two means percentage points, not a one-point-two-percent relative increase from an unstated baseline." Theo: [missing response] Hana: "My revised readout will show the center, both bounds, passing guardrails, and the unmet rollout condition. No launch approval or no-effect conclusion will be implied." Which response belongs in the gap?

A. Good. The practical threshold should be displayed on the same scale. Statistical uncertainty and business value are related in the decision, but they are not interchangeable labels.

A Correct. This is Theo's response in turn 18. It connects Hana's preceding point with Hana's follow-up. The practical threshold is the one-point business-relevant boundary used in the agreed lower-bound decision condition.

B Not here. This line belongs to turn 6, after Hana says: "Remind me of the actual rollout rule. I remember that we set a one-percentage-point threshold and also required the guardrails to pass." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Hana says: "That range includes a small negative effect and zero as well as useful positive outcomes. The current slide hides those possibilities by showing only the center." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Hana says: "The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement." It addresses that point, not the missing turn in the supplied exchange.

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Review 4 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: The role accountable for specified data decisions. Which term fits?

B. data owner

A Not here. indirect identifier: A field that may identify someone when combined with other information. The usual combination is "review indirect identifiers".

B Correct. data owner: The role accountable for specified data decisions. The usual combination is "obtain data-owner approval".

C Not here. data catalog: An organized inventory of datasets and their metadata. The usual combination is "consult the data catalog".

D Not here. disclosure review: Assessment of risks from releasing or sharing information. The usual combination is "conduct disclosure review".

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2. A colleague needs the term for this meaning: A large share of observations within a particular subgroup. Which term fits?

A. segment concentration

A Correct. segment concentration: A large share of observations within a particular subgroup. The usual combination is "report segment concentration".

B Not here. release timing: The timing of a product or system change. The usual combination is "compare release timing".

C Not here. insight: A decision-relevant interpretation supported by evidence. The usual combination is "state the insight".

D Not here. production change: A modification to a live operating system or service. The usual combination is "approve a production change".

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Review 4 / Explanations / continued

3. Rebuild the exchange from "A positive estimate is not the rollout rule". Hana: "Then should the slide say no effect? Zero is inside the interval, but so are positive effects. I do not want to replace one overclaim with another." Theo: [missing response] Hana: "Then the decision is clearer than the underlying effect. We know our agreed rollout rule failed, while the magnitude and even direction remain uncertain." Which response belongs in the gap?

A. No. The no-change value is included, but useful positive effects are included too. The readout does not establish exact zero or show that the two experiences are equivalent.

A Correct. This is Theo's response in turn 10. It connects Hana's preceding point with Hana's follow-up. Zero is the no-change value on the difference scale, but its presence in the interval does not prove the effect is exactly zero.

B Not here. This line belongs to turn 14, after Hana says: "Could we extend it until the lower bound clears one point? Or would that change the rule after we have already seen the result?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Hana says: "The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 16, after Hana says: "A new study might still be justified, but it would need a clear design and rationale rather than an attempt to rescue the original headline." It addresses that point, not the missing turn in the supplied exchange.

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Review 4 / Explanations / continued

4. Rebuild the exchange from "Five minutes to authorize the right investigation". Claudia: "Does approving this request also allow the analyst to change production if a likely cause appears? I want that distinction written clearly before I decide." Oren: [missing response] Claudia: "Then the decision is whether to allocate two days to a documented service issue, with the mobile concentration as a lead and the cause still open." Which response belongs in the gap?

D. The action boundary is investigation only. Any production change would need its separate evidence, review, and authority. We are not asking this meeting to bypass that process.

A Not here. This line belongs to turn 20, after Claudia says: "Then the decision is whether to allocate two days to a documented service issue, with the mobile concentration as a lead and the cause still open." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Claudia says: "We have five minutes and ten findings. Which decision needs my attention now, and what evidence supports it?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 10, after Claudia says: "But the concentration still gives the investigator a useful place to start. We can inspect the mobile paths without overstating what seventy out of eighty means." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Oren's response in turn 18. It connects Claudia's preceding point with Claudia's follow-up. The action boundary restricts the approval to investigation and readout, leaving production changes subject to their separate process.

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Additional case 1 / Explanations

Extended conversation

1. denominator

Turn 2. The denominator is the eligible visit count for the corresponding segment and period; those counts change substantially even though each period totals one thousand visits.

2. percentage points

Turn 4. Percentage points express the difference between the percentage values; both segment rates rise by two points, not by the same relative percentage.

3. mix shift

Turn 8. The mix shift moves a much larger share of visits into lower-converting segment A; changing weights can reverse the direction seen in the separate segment rates.

4. reference weights

Turn 12. The reference weights are the earlier ten-percent share for A and ninety-percent share for B; keeping them fixed creates a separate composition-standardized comparison.

5. standardization

Turn 16. Standardization applies a common weighting basis to improve comparability; it does not prove that an intervention changing traffic would leave every segment's rate unchanged.

6. causal effect

Turn 18. A causal effect concerns what an intervention changes; these aggregate and standardized comparisons alone do not isolate the effect of changing traffic allocation.

Return to the conversation, page 109

Transfer exchange

1. 50

Analyst: The earlier overall rate is 50 percent.

Turn 1. Twenty plus one hundred eighty equals two hundred conversions from four hundred visits, or fifty percent.

Additional case 1 / Explanations / continued

2. 40

Director: The later overall rate is 40 percent.

Turn 2. Ninety plus seventy equals one hundred sixty conversions from four hundred visits, or forty percent.

3. ten

Analyst: Each segment improved by ten percentage points.

Turn 3. A rises from twenty to thirty percent and B from sixty to seventy percent, each a ten-point increase.

4. composition

Director: Show the segment sizes so the changing composition is visible.

Turn 4. More visits belong to the lower-converting segment later, so the overall rate can fall despite both within-segment improvements.

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Additional case 2 / Explanations

Extended conversation

1. current record

Turn 2. The current record contains the customer's latest region, South; using it for every order reclassifies earlier sales instead of preserving their original regional attribution.

2. historical attribution

Turn 4. Historical attribution assigns each fact using the relevant state at its event time; it differs from regrouping all facts under today's customer attributes.

3. Type 2 dimension

Turn 6. A Type 2 dimension preserves successive versions of descriptive attributes so facts can be related to the version valid at the relevant time.

4. surrogate key

Turn 8. A surrogate key identifies the particular warehouse dimension row; the stable customer business identifier can appear on several historical versions.

5. time-based lookup

Turn 12. A time-based lookup matches the fact's relevant date to the dimension version's effective interval, rather than selecting the current version for every fact.

6. restatement

Turn 16. Restatement revises previously reported figures under a documented correction policy; the changed attribution and its reason must remain traceable.

Return to the conversation, page 114

Transfer exchange

1. East

Analyst: The April order belongs to East.

Turn 1. April precedes the May first move, so the order-date region is East.

Additional case 2 / Explanations / continued

2. West

Engineer: The June order belongs to West.

Turn 2. June follows the effective move date, making West the relevant historical version.

3. 700

Analyst: The unchanged grand total is 700 dollars.

Turn 3. Three hundred plus four hundred equals seven hundred under either allocation, so the total alone cannot verify the regional split.

4. keys

Engineer: Verify the effective-date intervals and the selected version keys.

Turn 4. Version keys must identify the dimension rows valid on each order date rather than merely the current customer row.

Return to the conversation, page 116

Additional case 3 / Explanations

Extended conversation

1. forecast origin

Turn 2. The forecast origin is the time when the prediction is issued; information generated afterward cannot be treated as an available input to that prediction.

2. data leakage

Turn 4. Data leakage introduces information unavailable for the intended prediction into model development or evaluation, making the reported performance unrepresentative of the real forecasting task.

3. backtest

Turn 8. A backtest evaluates predictions against historical outcomes while recreating the information and timing of the intended forecasting task.

4. holdout period

Turn 10. The holdout period is reserved for evaluation and was not used to fit or tune this model; repeatedly tuning against it would undermine that separation.

5. mean absolute error

Turn 12. Mean absolute error averages the magnitudes of prediction errors without cancellation; eighty tickets of total absolute error over ten days gives eight tickets per daily forecast.

6. baseline

Turn 14. The baseline is the agreed reference forecasting method evaluated on the same days and outcome definition; it gives the new model's error a meaningful comparison.

Return to the conversation, page 118

Transfer exchange

1. six

Analyst: The model mean absolute error is six tickets per forecast.

Turn 1. Forty-eight divided by eight gives six tickets of mean absolute error.

Additional case 3 / Explanations / continued

2. eight

Planner: The reference method averages eight tickets.

Turn 2. Sixty-four divided by eight gives eight tickets of mean absolute error.

3. twenty-five

Analyst: That is a twenty-five percent reduction in this error metric.

Turn 3. The reduction is two tickets relative to the eight-ticket reference error: two divided by eight equals twenty-five percent.

4. error

Planner: Do not relabel a six-ticket error as six-percent error.

Turn 4. The metric is in ticket units, and no percentage-error calculation is supplied.

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Find a phrase by purpose

Metric Definitions and Business Questions

Clarify the question: "Are we measuring sign-in or completed transactions?"

Define the counted entity: "Each user is counted once."

Name the event: "The qualifying event is a completed transaction."

Full phrase bank, page 7

Data Quality and Trust

Acknowledge the refresh: "The dashboard refresh completed at 09:00 UTC."

Separate source coverage: "That does not mean every source is current through Tuesday."

Name the lagging source: "Refunds are complete only through Monday."

Full phrase bank, page 19

Dashboards and Executive Reporting

Name the measure: "This is the share completed within 24 hours."

State the start: "The clock starts when the request is received."

Give count and base: "September recorded 480 timely completions out of 500."

Full phrase bank, page 32

Causation, Correlation, and Caveats

State the observed rates: "Attendees renewed at 90%; non-attendees at 80%."

Give the counts: "That is 90 of 100 versus 160 of 200."

Name the difference: "The observed gap is ten percentage points."

Full phrase bank, page 44

Find a phrase by purpose / continued

SQL, Models, and Transformation Logic

Ask about grain: "What does one row represent before the join?"

Name the key: "The tables match on order ID."

Describe the relationship: "O1 has two payment rows."

Full phrase bank, page 59

Experiment Readouts and Statistical Thinking

Report the center: "The estimated effect is plus 1.2 percentage points."

Show the uncertainty: "The 95% interval runs from minus 0.4 to plus 2.8 points."

Name the no-change value: "The interval includes zero."

Full phrase bank, page 70

Data Governance and Privacy

Define the purpose: "We are comparing weekly late-resolution rates by tier."

Link fields to the task: "The calculation needs counts by week and tier."

Limit identifiers: "Names and contact details are unnecessary for this question."

Full phrase bank, page 84

Insight Storytelling and Recommendations

Open with the action: "I recommend a two-day investigation of issue Q."

State the evidence: "The ticket rate rose from 2% to 8%."

Name the unit: "Those rates describe tickets, not unique users."

Full phrase bank, page 97

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References and publication notes

All cases, organizations, figures, and conversations are fictional teaching material, not transcripts. Communication practice does not replace qualified advice or current workplace procedures.

[1] UK Government. The Government Data Quality Framework.

Background for fit-for-purpose data, quality dimensions, and communicating limitations. The dashboard sources and freshness criteria here are fictional. Source checked 30 September 2026.

https://www.gov.uk/government/publications/the-government-data-quality-framework/the-government-data-quality-framework

[2] PostgreSQL Documentation. Table Expressions.

Background for join and grouping behavior. The order-and-payment examples are original, small teaching datasets rather than production queries. Source checked 30 September 2026.

https://www.postgresql.org/docs/current/queries-table-expressions.html

[3] NIST/SEMATECH e-Handbook. What Are Confidence Intervals?

Background for interval estimates and uncertainty language. The supplied experiment interval is fictional and is interpreted, not estimated from raw observations. Source checked 30 September 2026.

https://www.itl.nist.gov/div898/handbook/prc/section1/prc14.htm

[4] Information Commissioner's Office. Principle (c): Data Minimisation.

Background for purpose-linked data selection. This UK guidance is under review; the exercises use explicit fictional local rules, not universal legal conclusions. Source checked 30 September 2026.

https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-protection-principles/a-guide-to-the-data-protection-principles/data-minimisation/

Companion course

The website offers the eight-lesson course and related practice. This book can be used independently. No audio access or external account is needed for its exercises.

https://englishladder.com/efsp-data-analytics-business-intelligence.html

Define the measure. Make the finding useful.

Eight analytics conversations move from conflicting definitions and data-quality checks to trustworthy calculations, careful inference, and actionable recommendations.

Metric Definitions and Business Questions

900/2,000 is 45% of eligible users. 900/1,200 is 75% of users who logged in. Both calculations are valid for their stated bases, but only the first answers the supplied eligible-user question.

Data Quality and Trust

A refresh timestamp records when processing ran, not how current every input was. Even a latest event timestamp alone would not prove completeness. Here, the brief explicitly verifies different complete-through cutoffs for the two sources.

Dashboards and Executive Reporting

September is 96%: 480/500. August is 90%: 450/500. The increase is 6 percentage points, or approximately 6.67% relative to August. September exceeds the 95% target by 1 percentage point.

Causation, Correlation, and Caveats

The observed difference is 10 percentage points: 90%-80%. Relative to the non-attendee rate, it is 12.5%. Neither expression turns the voluntary comparison into an estimated causal benefit for mandatory training.

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